

Bounded solutions, L^p ($p > 1$) solutions and L^1 solutions for one dimensional BSDEs under general assumptions[☆]

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Received 26 February 2014; received in revised form 31 October 2015; accepted 28 November 2015

Available online 4 December 2015

Abstract

This paper aims at solving one dimensional backward stochastic differential equations (BSDEs) under weaker assumptions. We establish general existence, uniqueness, and comparison results for bounded solutions, L^p ($p > 1$) solutions and L^1 solutions of the BSDEs. The time horizon is allowed to be finite or infinite, and the generator g is allowed to have a general growth in y and a quadratic growth in z . As compensation, the generator g needs to satisfy a kind of one-sided linear or super-linear growth condition in y , instead of the monotonicity condition in y as is usually done. Many of our results improve considerably some known results, even though for the case of finite time horizon and the case of L^2 solution.

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MSC: 60H10

Keywords: Backward stochastic differential equation; Existence and uniqueness; Comparison theorem; L^p solution; Bounded solution; Infinite time interval

1. Preliminaries and introduction

Throughout this paper, let us fix a positive integer d and a positive time horizon T which can be finite or infinite. Moreover, let $\mathbb{R}_+ := [0, +\infty)$, $\mathbb{R}^+ := (0, +\infty)$ and let $L^1([0, T]; \mathbb{R}_+)$ and

[☆] Supported by the National Natural Science Foundation of China (No. 11371362), the China Postdoctoral Science Foundation (No. 2013M530173), the Qing Lan Project and the Fundamental Research Funds for the Central Universities (No. 2013RC20).

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$L^2([0, T]; \mathbb{R}_+)$, respectively, represent the set of nonnegative functions $u(\cdot)$ from $[0, T]$ to $\mathbb{R}_+$ such that

$$\int_0^T u(t) dt < +\infty \quad \text{and} \quad \int_0^T u^2(t) dt < +\infty.$$

Let $(\Omega, \mathcal{F}, P)$ be a complete probability space carrying a standard d -dimensional Brownian motion $(B_t)_{t \geq 0}$. We consider $(\mathcal{F}_t)_{t \geq 0}$ the natural filtration of $(B_t)_{t \geq 0}$ augmented by the P -null sets of $\mathcal{F}$ and we assume $\mathcal{F}_T = \mathcal{F}$. $\mathcal{P}$ denotes, as usual, the σ -algebra of predictable subsets of $\Omega \times [0, T]$. For each predictable subset A of $\Omega \times [0, T]$, let $\mathbb{1}_A$ equal to 1 when $(t, \omega) \in A$, and 0 otherwise. For every positive integer n , we use $|\cdot|$ to denote the norm of Euclidean space $\mathbb{R}^n$. For each $p > 0$, $L^p(\Omega, \mathcal{F}_T, P)$ represents the set of $\mathcal{F}_T$ -measurable random variables ξ such that $\mathbb{E}[|\xi|^p] < +\infty$, and $\mathcal{S}^p$ denotes the set of real-valued, $(\mathcal{F}_t)$ -adapted and continuous processes $(Y_t)_{t \in [0, T]}$ such that

$$\|Y\|_{\mathcal{S}^p} := \left(\mathbb{E} \left[\sup_{t \in [0, T]} |Y_t|^p \right] \right)^{1 \wedge 1/p} < +\infty.$$

If $p \geq 1$, $\|\cdot\|_{\mathcal{S}^p}$ is a norm on $\mathcal{S}^p$ and if $p \in (0, 1)$, $(X, X') \mapsto \|X - X'\|_{\mathcal{S}^p}$ defines a distance on $\mathcal{S}^p$. Under this metric, $\mathcal{S}^p$ is complete. Moreover, for each $p > 0$, let M^p denote the set of (equivalent classes of) $(\mathcal{F}_t)$ -progressively measurable, $\mathbb{R}^d$ -valued processes $(Z_t)_{t \in [0, T]}$ such that

$$\|Z\|_{M^p} := \left\{ \mathbb{E} \left[\left(\int_0^T |Z_t|^2 dt \right)^{p/2} \right] \right\}^{1 \wedge 1/p} < +\infty.$$

For $p \geq 1$, M^p is a Banach space endowed with this norm and for $p \in (0, 1)$, M^p is a complete metric space with the resulting distance. Finally, for each $p \geq 1$, we denote by $L^p(\Omega; L^1([0, T]; \mathbb{R}_+))$ the set of $(\mathcal{F}_t)$ -progressively measurable, nonnegative and real-valued process $(f_t)_{t \in [0, T]}$ such that

$$\mathbb{E} \left[\left(\int_0^T |f_t| dt \right)^p \right] < +\infty.$$

And, $L^1(\Omega; L^1([0, T]; \mathbb{R}_+))$ will be denoted simply by $L^1(\Omega \times [0, T])$.

We set $\mathcal{S} = \bigcup_{p \geq 1} \mathcal{S}^p$ and denote by $\mathcal{S}^\infty$ the set of predictable bounded processes, and by $L^\infty(\Omega, \mathcal{F}_T, P)$ the set of $\mathcal{F}_T$ -measurable bounded random variables. Let us recall that a continuous process $(Y_t)_{t \in [0, T]}$ belongs to the class (D) if the family $\{Y_\tau : \tau \in \Sigma_T\}$ is uniformly integrable, where and hereafter Σ_T stands for the set of all $(\mathcal{F}_t)$ -stopping times τ such that $\tau \leq T$.

In this paper, we consider the following one dimensional backward stochastic differential equation (BSDE in short for the remaining of this paper):

$$y_t = \xi + \int_t^T g(s, y_s, z_s) ds - \int_t^T z_s \cdot dB_s, \quad t \in [0, T], \quad (1)$$

where the random variable ξ is $\mathcal{F}_T$ -measurable, called the terminal condition of BSDE(1), the random function

$$g(\omega, t, y, z) : \Omega \times [0, T] \times \mathbb{R} \times \mathbb{R}^d \mapsto \mathbb{R}$$

is $\mathcal{P} \otimes \mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R}^d)$ measurable, called the generator of BSDE(1). We will sometimes use the notation $\text{BSDE}(\xi, g)$ to say that we consider the BSDE whose generator is g and whose terminal condition is ξ .

For convenience of the following discussion, we introduce the following definitions concerning the solutions of BSDE(1).

Definition 1.1. A solution of BSDE(1) is a pair of $(\mathcal{F}_t)$ -progressively measurable processes (y, z) with values in $\mathbb{R} \times \mathbb{R}^d$ such that dP -a.s., $t \mapsto y_t$ is continuous, $t \mapsto z_t$ belongs to $L^2(0, T)$, $t \mapsto g(t, y_t, z_t)$ belongs to $L^1(0, T)$, and dP -a.s., (1) holds true for each $t \in [0, T]$.

Definition 1.2. Assume that (y, z) is a solution of BSDE(1). If $(y, z) \in \mathcal{S}^\infty \times M^2$, then it is called a bounded solution; if $(y, z) \in \mathcal{S}^p \times M^p$ for some $p > 1$, then an L^p solution; if $(y, z) \in \mathcal{S}^\beta \times M^\beta$ for any $\beta \in (0, 1)$ and y belongs to the class (D), then an L^1 solution.

Definition 1.3. We say that (y, z) is a maximal bounded (resp. L^p ($p > 1$) and L^1) solution of BSDE(1) if it is a bounded (resp. L^p and L^1) solution and $y \geq y'$ for any bounded (resp. L^p and L^1) solution (y', z') . Similarly, if $y \leq y'$, then (y, z) is called a minimal bounded (resp. L^p and L^1) solution.

Since the first existence and uniqueness result for nonlinear multidimensional BSDEs with square integrable parameters was introduced by Pardoux and Peng [32] under the Lipschitz assumption of g , BSDEs have been extensively studied, and many applications have been found in mathematical finance, stochastic control, partial differential equations and so on (see [2,4,12–14,22,26,30,31,33–37] for details).

From the beginning, many authors attempted to improve the result of [32] by weakening the Lipschitz hypothesis on g , see [1,2,4–8,12–21,23,24,26–31,33,38,39], or the L^2 integrability assumptions on ξ , see [5,6,10,14,16,17,20,34,38,39], or relaxing the finite time horizon T to a stopping time or infinity, see [11,17,19,30,31]. From these results it is not difficult to see that the case of one dimensional BSDEs is easy to handle due to the presence of the comparison theorem of solutions, see [6,7,9–11,14–16,19,20,23,24,26–29,34,38,39].

This paper aims at solving one dimensional BSDEs under weaker assumptions. We establish general existence, uniqueness, and comparison results for bounded solutions, L^p ($p > 1$) solutions and L^1 solutions of the BSDEs. The time horizon T is allowed to be finite or infinite, and the generator g is allowed to have a general growth in y and a quadratic growth in z . As compensation, the generator g needs to satisfy a kind of one-sided linear or super-linear growth condition in y , instead of the monotonicity condition in y as is usually done. Many of our results improve considerably some known results, even though for the case of finite time horizon and the case of L^2 solution.

More specifically, with respect to the existence for bounded solutions, L^p ($p > 1$) solutions and L^1 solutions of BSDEs, we would like to list respectively several existing results and our results as follows in order to compare with each other. Here, we always assume that the generator g is continuous in (y, z) , and T is a finite real number.

First, when ξ is bounded there exists a maximal (resp. minimal) bounded solution of $\text{BSDE}(\xi, g)$ under one of the following two groups of conditions:

- g has a super-linear growth in y and a quadratic growth in z , i.e., there exist a constant $C > 0$ and a continuous function $l : \mathbb{R} \mapsto \mathbb{R}^+$ such that

$$|g(\omega, t, y, z)| \leq l(y) + C|z|^2,$$

where $\int_0^{+\infty} \frac{dx}{l(x)} = \int_{-\infty}^0 \frac{dx}{l(x)} = +\infty$

(see Lepeltier and San Martín [28] and Kobylanski [26]).

- There exist two constants $\alpha > 0$ and $\beta > 0$ together with a convex $\mathcal{C}^1$ function $\rho : \mathbb{R}_+ \mapsto \mathbb{R}_+$ with $\rho(0) = 0$ and a continuous function $\varphi : \mathbb{R}_+ \mapsto \mathbb{R}_+$ with $\varphi(0) = 0$ such that

$$(i) (g(\omega, t, y, z) - g(\omega, t, 0, z)) \operatorname{sgn}(y) \leq \rho(|y|),$$

$$(ii) |g(\omega, t, y, z)| \leq \alpha + \varphi(|y|) + \beta|z|^2,$$

where $\int_0^{+\infty} \frac{ds}{\rho(s)+\varphi} = +\infty$

(see Briand and Hu [7] and Briand, Lepeltier and San Martín [8]).

It is not hard to verify that neither of the above two groups of conditions is satisfied for the following two generators:

$$g_1(\omega, t, y, z) := |z|^2 e^y + y \cos y \quad \text{and} \quad g_2(\omega, t, y, z) := -y^3 + |z|^{\frac{3}{2}} \sin y.$$

However, they both satisfy the following condition:

- g has a one-sided super-linear growth in y and a quadratic growth in z , i.e., there exist two continuous functions $l : \mathbb{R} \mapsto \mathbb{R}^+$ and $\varphi : \mathbb{R} \mapsto \mathbb{R}_+$ such that

$$(i) g(\omega, t, y, z) \operatorname{sgn}(y) \leq l(y) + \varphi(y)|z|^2,$$

$$(ii) |g(\omega, t, y, z)| \leq \varphi(y)(1 + |z|^2),$$

where $\int_0^{+\infty} \frac{dx}{l(x)} = \int_{-\infty}^0 \frac{dx}{l(x)} = +\infty$.

Then, by Theorem 3.1 of this paper we know that when ξ is bounded there exists a maximal (resp. minimal) bounded solution of BSDE(ξ, g_i) for $i = 1, 2$. In addition, the case of $T = +\infty$ is also considered in Theorem 3.1.

Second, when $\xi \in L^p(\Omega, \mathcal{F}_T, P)$ for some $p > 1$ there exists a maximal (resp. minimal) L^p solution of BSDE(ξ, g) under one of the following two groups of conditions:

- g has a linear growth in (y, z) , i.e., there exists a constant $C > 0$ such that

$$|g(\omega, t, y, z)| \leq C(1 + |y| + |z|)$$

(see Lepeltier and San Martín [27] and Chen [10]).

- There exist two constants $\mu \in \mathbb{R}$ and $A > 0$ together with a continuous adapted process $g_t \in L^p(\Omega; L^1([0, T]; \mathbb{R}_+))$ and a continuous function $\varphi : \mathbb{R}_+ \mapsto \mathbb{R}_+$ with $\varphi(0) = 0$ such that

$$(i) (g(\omega, t, y_1, z) - g(\omega, t, y_2, z)) \operatorname{sgn}(y_1 - y_2) \leq \mu|y_1 - y_2|,$$

$$(ii) |g(\omega, t, y, z)| \leq g_t(\omega) + \varphi(|y|) + A|z|$$

(see Briand, Lepeltier and San Martín [8]).

It is not hard to verify that neither of the above two groups of conditions is satisfied for the following two generators:

$$g_1(\omega, t, y, z) := |z|^2(1 - e^y) + |z| \sin |z| \quad \text{and} \quad g_2(\omega, t, y, z) := -y^5 + \cos(y|z|).$$

However, they both satisfy the following condition:

- g has a one-sided linear growth in (y, z) and a quadratic growth in z , i.e., there exist a constant $C > 0$ and a continuous function $\varphi : \mathbb{R} \mapsto \mathbb{R}_+$ such that

$$(i) g(\omega, t, y, z) \operatorname{sgn}(y) \leq C(1 + |y| + |z|),$$

$$(ii) |g(\omega, t, y, z)| \leq \varphi(y)(1 + |z|^2).$$

Then, by Theorem 5.1 of this paper we know that when $\xi \in L^p(\Omega, \mathcal{F}_T, P)$ for some $p > 1$ there exists an L^p solution of BSDE(ξ, g_i) for $i = 1, 2$. In addition, the case of $T = +\infty$ is also considered in Theorem 5.1, and the existence of a maximal (resp. minimal) L^p solution is further investigated in Theorem 5.2.

Third, when $\xi \in L^1(\Omega, \mathcal{F}_T, P)$ there exists an L^1 solution of BSDE(ξ, g) under one of the following two groups of conditions:

- g has a linear growth in y and a sub-linear growth in z , i.e., there exist two constants $C > 0$ and $\alpha \in (0, 1)$ such that

$$|g(\omega, t, y, z)| \leq C(1 + |y| + |z|^\alpha)$$

(see the first version of Briand and Hu [6]).

- There exist constants $\mu \in \mathbb{R}$, $\lambda \geq 0$, $\delta \geq 0$ and $\alpha \in (0, 1)$ together with an $(\mathcal{F}_t)$ -progressively measurable nonnegative process $g_t \in L^1(\Omega \times [0, T])$ such that

$$(i) (g(\omega, t, y_1, z) - g(\omega, t, y_2, z)) \operatorname{sgn}(y_1 - y_2) \leq \mu |y_1 - y_2|,$$

$$(ii) |g(\omega, t, y, z_1) - g(\omega, t, y, z_2)| \leq \lambda |z_1 - z_2|,$$

$$(iii) \text{ for each } r \geq 0, \psi_r(t) := \sup_{|y| \leq r} |g(\omega, t, y, 0)| \in L^1(\Omega \times [0, T]),$$

$$(iv) |g(\omega, t, y, z) - g(\omega, t, y, 0)| \leq \delta(g_t(\omega) + |y| + |z|)^\alpha$$

(see Briand, Delyon, Hu, Pardoux and Stoica [5] in the multidimensional case).

It is not hard to verify that neither of the above two groups of conditions is satisfied for the following two generators:

$$g_1(\omega, t, y, z) := -|z|^2 y^3 + \sqrt[3]{|z|} \quad \text{and} \quad g_2(\omega, t, y, z) := e^{-y} \sqrt{|z|} + \sqrt{1 + |y| + |z|}.$$

However, they both satisfy the following condition:

- g has a one-sided linear growth in y , a one-sided sub-linear growth in z and a quadratic growth in z , i.e., there exist two constants $C > 0$, $\alpha \in (0, 1)$ and a continuous function $\varphi : \mathbb{R} \mapsto \mathbb{R}_+$ such that

$$(i) g(\omega, t, y, z) \operatorname{sgn}(y) \leq C(1 + |y| + |z|^\alpha),$$

$$(ii) |g(\omega, t, y, z)| \leq \varphi(y)(1 + |z|^2).$$

Then, by Theorem 6.1 of this paper we know that when $\xi \in L^1(\Omega, \mathcal{F}_T, P)$ there exists an L^1 solution of BSDE(ξ, g_i) for $i = 1, 2$. In addition, the case of $T = +\infty$ is also considered in Theorem 6.1, and the existence of a maximal (resp. minimal) L^1 solution is further investigated in Theorem 6.2.

In the sequel, with respect to the uniqueness and comparison results for bounded solutions, L^p ($p > 1$) solutions and L^1 solutions of BSDEs, we also list respectively several existing results and our results as follows.

Briand and Hu [7] established a comparison theorem for bounded solutions of BSDEs with finite time horizon when one of the generators is Lipschitz in y and concave or convex in z ; Morlais [30] obtained a comparison theorem for bounded solutions of BSDEs with infinite time horizon when one of the generators satisfies a monotonicity condition in y and a local Lipschitz

condition in z . Under the conditions that one of the generators only satisfies a one-sided Osgood condition in y (see assumption (2A1) and Remark 2.1 in Section 2 for details), and a local Lipschitz condition (resp. a concavity or convexity condition) in z , Theorems 4.1 and 4.2 of this paper respectively prove a comparison theorem for bounded solutions of BSDEs with finite or infinite time horizon, extending two results mentioned above.

Fan, Jiang and Tian [19] established a comparison theorem for L^2 solutions of BSDEs with finite or infinite time horizon when one of the generators satisfies a one-sided Osgood condition in y and a uniform continuity condition in z , which generalizes four classical comparison theorems obtained respectively in El Karoui, Peng and Quenez [14], Cao and Yan [9], Chen and Wang [11] and Briand and Hu [6]. Theorem 2.1 of this paper further extends this result to the case of the L^p ($p > 1$) solution. More importantly, in this paper we eliminate the concavity condition with respect to function $\rho(\cdot)$ in the one-sided Osgood condition.

To the best of our knowledge, Briand and Hu [6] first put forward and proved a comparison theorem for L^1 solutions of BSDEs with finite time horizon when one of the generators satisfies a monotonicity condition in y and a Lipschitz condition together with a sub-linear growth condition in z . Recently, Fan and Liu [20], Xiao, Li and Fan [39], Fan and Jiang [16] and Tian, Jiang and Shi [38] further establish the comparison results for L^1 solutions of BSDEs with finite time horizon under the conditions that one of the generators satisfies a monotonicity condition or a Osgood condition in y and a quasi-Hölder continuity condition in z . Theorem 2.4 of this paper unifies these results to the case of BSDEs with finite or infinite time horizon when one of the generators satisfies a one-sided Osgood condition in y and a uniform continuity condition together with a sub-linear growth condition in z . For example, the following generator g does not satisfy their conditions but satisfies our conditions:

$$g(\omega, t, y, z) = \begin{cases} \sqrt{|z|}, & 0 \leq |z| \leq 1; \\ (n-1)^4, & (n-1)^4 < |z| \leq n^4 - 2n + 1; \\ |z| + n^2 - n^4, & n^4 - 2n + 1 < |z| \leq n^4, \end{cases} \quad n = 2, 3, 4, \dots$$

More specifically, this generator g satisfies the uniform continuity condition as well as the sub-linear growth condition in z , but it does not satisfy the quasi-Hölder continuity condition in z .

Finally, we would like to mention that several new comparison results for the maximal and minimal solutions of BSDEs are put forward and proved (see, for example, Theorems 2.2–2.3, 3.2–3.3, 5.3–5.4 and 6.3–6.4), and they also play an important role in the proof of our existence results. Furthermore, by products, three general existence and uniqueness results for bounded solutions, L^p ($p > 1$) solutions and L^1 solutions of BSDEs are also obtained respectively (see Theorems 4.3, 5.5 and 6.5). In addition, we also point out that our results are all obtained due to the application of new ideas and new techniques or the development of those existing ideas and methods.

The remaining of this paper is organized as follows. In Section 2, we establish several comparison theorems for the (maximal and minimal) L^p ($p > 1$) solutions and the L^1 solutions of BSDEs. Then, we prove an existence result together with two comparison results for the maximal and minimal bounded solutions in Section 3, and two comparison theorems together with an existence and uniqueness result for the bounded solutions in Section 4. Finally, in Sections 5 and 6, we establish several existence results, comparison theorems, and existence and uniqueness results for the (maximal and minimal) L^p ($p > 1$) and L^1 solutions of BSDEs respectively.

2. Comparison theorems of L^p ($p > 1$) solutions and L^1 solutions

In this section, we will establish several comparison theorems for the (maximal and minimal) L^p ($p > 1$) solutions and the L^1 solutions of BSDEs. Let us first introduce the following assumptions on the generator g , where we assume that $0 < T \leq +\infty$:

(2A1) There exist a function $u(\cdot) \in L^1([0, T]; \mathbb{R}_+)$ and a nondecreasing continuous function $\rho(\cdot) : \mathbb{R}_+ \mapsto \mathbb{R}_+$ with linear growth such that $dP \times dt$ -a.e.,

$$(g(\omega, t, y_1, z) - g(\omega, t, y_2, z)) \operatorname{sgn}(y_1 - y_2) \leq u(t)\rho(|y_1 - y_2|), \quad \forall y_1, y_2, z.$$

Assume further that $\rho(0) = 0$, $\rho(u) > 0$ for $u > 0$ and $\int_0^+ \frac{1}{\rho(u)} du = +\infty$.

(2A2) There exist a function $v(\cdot) \in L^2([0, T]; \mathbb{R}_+)$ and a nondecreasing continuous function $\phi(\cdot) : \mathbb{R}_+ \mapsto \mathbb{R}_+$ with $\phi(0) = 0$ such that $dP \times dt$ -a.e.,

$$|g(\omega, t, y, z_1) - g(\omega, t, y, z_2)| \leq v(t)\phi(|z_1 - z_2|), \quad \forall y, z_1, z_2.$$

Without loss of generality, here and henceforth we can always assume that for all $x \in \mathbb{R}_+$, $0 \leq \phi(x) \leq ax + b$. Furthermore, we assume also that $v(t) \in L^1([0, T]; \mathbb{R}_+)$ in the case where $b \neq 0$.

(2A3) $dP \times dt$ -a.e., $g(\omega, t, \cdot, \cdot) : \mathbb{R} \times \mathbb{R}^d \mapsto \mathbb{R}$ is continuous.

(2A4) There exist two functions $u(\cdot) \in L^1([0, T]; \mathbb{R}_+)$, $v(\cdot) \in L^2([0, T]; \mathbb{R}_+)$ and a process $f_t \in L^2(\Omega; L^1([0, T]; \mathbb{R}_+))$ such that $dP \times dt$ -a.e.,

$$|g(\omega, t, y, z)| \leq f_t(\omega) + u(t)|y| + v(t)|z|, \quad \forall y, z.$$

(2A5) There exist a constant $\alpha \in (0, 1)$, a function $\lambda(\cdot) : [0, T] \mapsto \mathbb{R}_+$ and an $(\mathcal{F}_t)$ -progressively measurable nonnegative process $(f_t)_{t \in [0, T]} \in L^1([0, T] \times \Omega)$ such that $dP \times dt$ -a.e.,

$$|g(\omega, t, y, z) - g(\omega, t, y, 0)| \leq \lambda(t)(f_t(\omega) + |y| + |z|)^\alpha, \quad \forall y, z.$$

We also assume that

$$\int_0^T \left(\lambda(t) + \lambda^{\frac{1}{1-\alpha}}(t) \right) dt < +\infty.$$

(2A5') There exist a constant $\alpha \in (0, 1)$ and a function $\lambda(\cdot) : [0, T] \mapsto \mathbb{R}_+$ such that $dP \times dt$ -a.e.,

$$|g(\omega, t, y, z) - g(\omega, t, y, 0)| \leq \lambda(t)|z|^\alpha, \quad \forall y, z.$$

We also assume that

$$\int_0^T \lambda^{\frac{2}{2-\alpha}}(t) dt < +\infty.$$

Remark 2.1. In (2A1), we do not assume that $\rho(\cdot)$ is a concave function required by Theorem 2 of Fan, Jiang and Tian [19].

The following lemma will play an important role in the proof of main results of this paper.

Lemma 2.1. Assume that $0 < T \leq +\infty$, $\{b_n\}_{n=1}^{+\infty}$ is a nonnegative and non-increasing real sequence, $\beta(t) \in L^1([0, T]; \mathbb{R}_+)$ and $\psi(\cdot) : \mathbb{R}_+ \mapsto \mathbb{R}_+$ is a nondecreasing continuous function

with linear growth. Let $\{(u_n(t))_{t \in [0, T]}\}_{n=1}^{+\infty}$ be a sequence of non-negative $(\mathcal{F}_t)$ -progressively measurable and continuous processes satisfying

$$\mathbb{E}_n \left[\sup_{t \in [0, T]} u_n(t) \right] < +\infty$$

and for each $t \in [0, T]$,

$$u_n(t) \leq b_n + \mathbb{E}_n \left[\int_t^T \beta(s) \psi(u_n(s)) \, ds \middle| \mathcal{F}_t \right] \quad dP\text{-a.s.}, \quad (2)$$

where $\mathbb{E}_n[X|\mathcal{F}_t]$ represents the conditional expectation of random variable X with respect to $\mathcal{F}_t$ under a probability measure P_n defined on $(\Omega, \mathcal{F}_T)$, which is equivalent to P and may depend on n . If $\lim_{n \rightarrow \infty} b_n = 0$, $\psi(0) = 0$, $\psi(u) > 0$ for $u > 0$, and

$$\int_{0+} \frac{1}{\psi(u)} \, du = +\infty, \quad (3)$$

then for each $t \in [0, T]$,

$$\lim_{n \rightarrow \infty} u_n(t) = 0 \quad dP\text{-a.s.} \quad (4)$$

Proof. Since ψ is of linear growth, we can get the existence of a constant k such that $\psi(x) \leq k(1+x)$ for all $x \in \mathbb{R}_+$. Then, in view of $b_n \leq b_1$, $\beta(\cdot) \in L^1([0, T]; \mathbb{R}_+)$ and Fubini's Theorem, by (2) we can obtain that for each $t \in [0, T]$, dP -a.s.,

$$\mathbb{E}_n[u_n(r)|\mathcal{F}_t] \leq b_1 + k \int_0^T \beta(s) \, ds + k \int_r^T \beta(s) \mathbb{E}_n[u_n(s)|\mathcal{F}_t] \, ds, \quad r \in [t, T].$$

Thus, Gronwall's inequality yields that for each $t \in [0, T]$, dP -a.s.,

$$\mathbb{E}_n[u_n(r)|\mathcal{F}_t] \leq \left(b_1 + k \int_0^T \beta(s) \, ds \right) e^{k \int_r^T \beta(s) \, ds}, \quad r \in [t, T].$$

Letting $r = t$ in the above inequality we get that for each $t \in [0, T]$,

$$\sup_{n \geq 1} u_n(t) \leq C := \left(b_1 + k \int_0^T \beta(s) \, ds \right) e^{k \int_0^T \beta(s) \, ds} \quad dP\text{-a.s.} \quad (5)$$

In the sequel, in view of the linear growth of ψ , for each $n \geq 1$ we can define the function $\psi_n : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ as follows

$$\psi_n(x) = \sup_{y \in \mathbb{R}_+} \{ \psi(y) - (n + 2k)|x - y| \}.$$

It is well known that ψ_n is well defined and Lipschitz. Moreover, the sequence $\{\psi_n\}_{n=1}^{+\infty}$ is non-increasing and converges to ψ . Thus, for each $n \geq 1$, noticing that $\beta(\cdot) \in L^1([0, T]; \mathbb{R}_+)$, we can let $v_n : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ be the solution of the following backward ordinary differential equation (ODE for short):

$$v_n(t) = b_n + \int_t^T \beta(s) \psi_n(v_n(s)) \, ds, \quad t \in [0, T].$$

Since $\{\psi_n\}_{n=1}^\infty$ and $\{b_n\}_{n=1}^\infty$ are both non-increasing sequences, we know that $v_{n+1} \leq v_n$ for each $n \geq 1$. This implies that, in view of the fact that $\lim_{n \rightarrow \infty} b_n = 0$ and $\{\psi_n\}_{n=1}^\infty$ converges to ψ as $n \rightarrow \infty$, the sequence $\{v_n\}_{n=1}^\infty$ converges pointwise to a function $v : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ which satisfies

$$v(t) = \int_t^T \beta(s) \psi(v(s)) ds, \quad t \in [0, T].$$

In view of (3) and the fact that $\beta(\cdot) \in L^1([0, T]; \mathbb{R}_+)$, Bihari's inequality (see Bihari [3] for details) yields that $v(t) = 0$ for each $t \in [0, T]$.

Now for $n, j \geq 1$, let v_n^j be the function defined recursively as follows:

$$\begin{aligned} v_n^1(t) &\equiv C, \\ v_n^{j+1}(t) &= b_n + \int_t^T \beta(s) \psi_n(v_n^j(s)) ds, \quad j \geq 1, \quad t \in [0, T], \end{aligned} \quad (6)$$

where C is defined in (5). Since ψ_n is Lipschitz and $\beta(\cdot) \in L^1([0, T]; \mathbb{R}_+)$, we know that $v_n^j \rightarrow v_n$ as $j \rightarrow \infty$. On the other hand, it is easily seen by induction that for all $n, j \geq 1$ and each $t \in [0, T]$,

$$u_n(t) \leq v_n^j(t) \quad dP\text{-a.s.} \quad (7)$$

Indeed, for $j = 1$ the formula holds true by (5). Suppose it also holds for some j , then

$$dP\text{-a.s.}, \quad \psi(u_n(s)) \leq \psi(v_n^j(s)) \leq \psi_n(v_n^j(s)), \quad s \in [0, T].$$

In view of (2), the previous inequality and (6), we can deduce that for all $n \geq 1$ and each $t \in [0, T]$,

$$u_n(t) \leq v_n^{j+1}(t) \quad dP\text{-a.s.}$$

Thus, (7) follows. Finally, taking limit in (7) as first $j \rightarrow \infty$, and then $n \rightarrow \infty$, we obtain (4). The proof of Lemma 2.1 is then completed. $\square$

The following Theorem 2.1 establishes a general comparison theorem for L^p ($p > 1$) solutions of BSDEs. It can be regarded as a generalization of Theorem 2 in Fan, Jiang and Tian [19], where only L^2 solutions are considered, and the concavity condition of $\rho(\cdot)$ in (2A1) is also required.

Theorem 2.1. Assume that $0 < T \leq +\infty$, ξ and ξ' are two $\mathcal{F}_T$ -measurable random variables, g and g' are two generators of BSDEs and (y, z) and (y', z') are, respectively, a solution of BSDE(ξ, g) and BSDE(ξ', g'). Assume further that $(y - y')^+ \in \mathcal{S}$. If $\xi \leq \xi'$ dP -a.s. and one of the following two statements holds true:

(i) g satisfies (2A1) and (2A2), and

$$\mathbb{1}_{y_t > y'_t} (g(t, y'_t, z'_t) - g'(t, y'_t, z'_t)) \leq 0 \quad dP \times dt\text{-a.e.}; \quad (8)$$

(ii) g' satisfies (2A1) and (2A2), and

$$\mathbb{1}_{y_t > y'_t} (g(t, y_t, z_t) - g'(t, y_t, z_t)) \leq 0 \quad dP \times dt\text{-a.e.}, \quad (9)$$

then for each $t \in [0, T]$, we have

$$y_t \leq y'_t \quad dP\text{-a.s.}$$

Proof. We will only prove the case (i). The other case can be proved in the same way. Let us fix $k \in \mathbb{N}^*$ and denote τ_k the stopping time

$$\tau_k = \inf \left\{ t \in [0, T] : \int_0^t (|z_s|^2 + |z'_s|^2) \, ds \geq k \right\} \wedge T.$$

Tanaka's formula leads to the equation, setting $\hat{y}_t = y_t - y'_t$, $\hat{z}_t = z_t - z'_t$,

$$\hat{y}_{t \wedge \tau_k}^+ \leq \hat{y}_{\tau_k}^+ + \int_{t \wedge \tau_k}^{\tau_k} \mathbb{1}_{\hat{y}_s > 0} (g(s, y_s, z_s) - g'(s, y'_s, z'_s)) \, ds - \int_{t \wedge \tau_k}^{\tau_k} \mathbb{1}_{\hat{y}_s > 0} \hat{z}_s \cdot dB_s. \quad (10)$$

First of all, since $\mathbb{1}_{\hat{y}_s > 0} (g(s, y'_s, z'_s) - g'(s, y'_s, z'_s))$ is non-positive, we have

$$\begin{aligned} & \mathbb{1}_{\hat{y}_s > 0} (g(s, y_s, z_s) - g'(s, y'_s, z'_s)) \\ &= \mathbb{1}_{\hat{y}_s > 0} (g(s, y_s, z_s) - g(s, y'_s, z'_s)) + \mathbb{1}_{\hat{y}_s > 0} (g(s, y'_s, z'_s) - g'(s, y'_s, z'_s)) \\ &\leq \mathbb{1}_{\hat{y}_s > 0} (g(s, y_s, z_s) - g(s, y'_s, z'_s)) + \mathbb{1}_{\hat{y}_s > 0} (g(s, y'_s, z'_s) - g'(s, y'_s, z'_s)) \end{aligned}$$

and we deduce, using the assumptions (2A1) and (2A2) for g , that

$$\mathbb{1}_{\hat{y}_s > 0} (g(s, y_s, z_s) - g'(s, y'_s, z'_s)) \leq u(s) \rho(\hat{y}_s^+) + \mathbb{1}_{\hat{y}_s > 0} v(s) \phi(|\hat{z}_s|). \quad (11)$$

From the proof of Theorem 1 in Fan, Jiang and Davison [18] we know that, with $c = a + b$,

$$\phi(x) \leq (n + 2c)x + \mathbf{1}_{b \neq 0} \phi\left(\frac{2c}{n + 2c}\right), \quad \forall x \in \mathbb{R}_+, \quad \forall n \geq 1. \quad (12)$$

Combining (10)–(12) yields that for each $n \geq 1$ and each $t \in [0, T]$,

$$\begin{aligned} \hat{y}_{t \wedge \tau_k}^+ &\leq a_n + \hat{y}_{\tau_k}^+ + \int_{t \wedge \tau_k}^{\tau_k} [u(s) \rho(\hat{y}_s^+) + \mathbb{1}_{\hat{y}_s > 0} (n + 2c) v(s) |\hat{z}_s|] \, ds - \int_{t \wedge \tau_k}^{\tau_k} \mathbb{1}_{\hat{y}_s > 0} \hat{z}_s \cdot dB_s \\ &= a_n + \hat{y}_{\tau_k}^+ + \int_{t \wedge \tau_k}^{\tau_k} u(s) \rho(\hat{y}_s^+) \, ds - \int_{t \wedge \tau_k}^{\tau_k} \mathbb{1}_{\hat{y}_s > 0} \hat{z}_s \\ &\quad \cdot \left[-\frac{(n + 2c)v(s)\hat{z}_s}{|\hat{z}_s|} \mathbb{1}_{|\hat{z}_s| \neq 0} \, ds + dB_s \right], \end{aligned} \quad (13)$$

where

$$a_n = \mathbf{1}_{b \neq 0} \phi\left(\frac{2c}{n + 2c}\right) \cdot \int_0^T v(s) \, ds \rightarrow 0 \quad \text{by (2A2) as } n \rightarrow \infty. \quad (14)$$

Let P_n be the probability on $(\Omega, \mathcal{F}_T)$ which is equivalent to P and defined by

$$\frac{dP_n}{dP} := \exp \left\{ (n + 2c) \int_0^T \frac{v(s)\hat{z}_s}{|\hat{z}_s|} \mathbb{1}_{|\hat{z}_s| \neq 0} \cdot dB_s - \frac{1}{2} (n + 2c)^2 \int_0^T \mathbb{1}_{|\hat{z}_s| \neq 0} v^2(s) \, ds \right\}.$$

It is worth noticing that dP_n/dP has moments of all orders since $v(\cdot) \in L^2([0, T]; \mathbb{R}_+)$. By Girsanov's theorem, under P_n the process

$$B_n(t) = B_t - \int_0^t \frac{(n + 2c)v(s)\hat{z}_s}{|\hat{z}_s|} \mathbb{1}_{|\hat{z}_s| \neq 0} \, ds, \quad t \in [0, T],$$

is an $(\mathcal{F}_t, P_n)$ -Brownian motion. Moreover, the process

$$\left(\int_0^{t \wedge \tau_k} \mathbb{1}_{\hat{y}_s > 0} \hat{z}_s \cdot dB_n(s) \right)_{0 \leq t \leq T}$$

is an $(\mathcal{F}_t, P_n)$ -martingale. Let $\mathbb{E}_n[X|\mathcal{F}_t]$ represent the conditional expectation of the random variable X with respect to $\mathcal{F}_t$ under P_n . By taking the conditional expectation with respect to $\mathcal{F}_t$ under P_n in (13) we get that for each $n \geq 1$ and $t \in [0, T]$,

$$\hat{y}_{t \wedge \tau_k}^+ \leq a_n + \mathbb{E}_n[\hat{y}_{\tau_k}^+ | \mathcal{F}_t] + \mathbb{E}_n \left[\int_{t \wedge \tau_k}^{\tau_k} u(s) \rho(\hat{y}_s^+) ds \middle| \mathcal{F}_t \right] \quad dP\text{-a.s.}$$

Furthermore, in view of the facts that $\tau_k \rightarrow T$ as $k \rightarrow \infty$, $(\hat{y})^+$ belongs to $\mathcal{S}$, $\xi \leq \xi'$ and $u(\cdot) \in L^1([0, T]; \mathbb{R}_+)$, letting $k \rightarrow \infty$ in the above inequality and using Lebesgue's dominated convergence theorem yields that for each $t \in [0, T]$,

$$\hat{y}_t^+ \leq a_n + \mathbb{E}_n \left[\int_t^T u(s) \rho(\hat{y}_s^+) ds \middle| \mathcal{F}_t \right] \quad dP\text{-a.s.}$$

Thus, in view of (14), applying Lemma 2.1 with $u_n(t) \equiv \hat{y}_t^+$, $b_n = a_n$, $\beta(s) = u(s)$ and $\psi(u) = \rho(u)$ yields that for each $t \in [0, T]$,

$$\hat{y}_t^+ = \lim_{n \rightarrow \infty} \hat{y}_t^+ = \lim_{n \rightarrow \infty} u_n(t) = 0 \quad dP\text{-a.s.}$$

That is to say, for each $t \in [0, T]$, $y_t \leq y'_t$ dP -a.s. Theorem 2.1 is then proved. $\square$

It should be especially noted that the presence of the indicator function makes that (8) and (9) can be more easily satisfied than the usual form. This important observation is one of the start points of this paper.

The following Remark 2.2 gives an easily verifiable condition to ensure that (8) or (9) holds true, which will be used several times and play an important role later.

Remark 2.2. Assume that c is a constant. If

$$dP\text{-a.s.}, \quad \forall t \in [0, T], \quad y_t \leq c$$

and

$$dP \times dt\text{-a.e.}, \quad \forall y < c, \quad \forall z \in \mathbb{R}^d, \quad g(t, y, z) \leq g'(t, y, z),$$

then (8) holds true. Similarly, if

$$dP\text{-a.s.}, \quad \forall t \in [0, T], \quad y'_t \geq c$$

and

$$dP \times dt\text{-a.e.}, \quad \forall y > c, \quad \forall z \in \mathbb{R}^d, \quad g(t, y, z) \leq g'(t, y, z),$$

then (9) holds true.

Let us further introduce the following Lemma 2.2, which comes from Theorem 1 in Fan, Jiang and Tian [19]. It will be generalized in Section 5.

Lemma 2.2. Assume that $0 < T \leq +\infty$ and the generator g satisfies (2A3) and (2A4). Then for each $\xi \in L^2(\Omega, \mathcal{F}_T, P)$, BSDE(ξ, g) has both a minimal L^2 solution and a maximal L^2 solution.

The following Theorem 2.2 establishes a general comparison theorem for the maximal L^2 solutions, which will be also generalized in Section 5.

Theorem 2.2. Assume that $0 < T \leq +\infty$, $\xi, \xi' \in L^2(\Omega, \mathcal{F}_T, P)$, g and g' are two generators of BSDEs, and (y, z) is any L^2 solution of BSDE(ξ, g). Assume further that g' satisfies (2A3) and (2A4), and (y', z') is the maximal L^2 solution of BSDE(ξ', g') (recall Lemma 2.2). If $\xi \leq \xi'$ dP-a.s. and (9) holds true, then for each $t \in [0, T]$,

$$y_t \leq y'_t \quad \text{dP-a.s.}$$

Proof. Since g' satisfies (2A3) and (2A4), by the proof of Theorem 1 in Fan, Jiang and Tian [19] we know that if we take for each $n \geq 1$ and $(\omega, t, y, z) \in \Omega \times [0, T] \times \mathbb{R} \times \mathbb{R}^d$,

$${}^n g'(\omega, t, y, z) := \sup_{(u, v) \in \mathbb{R}^{1+d}} \{g'(\omega, t, u, v) - nu(t)|y - u| - nv(t)|z - v|\},$$

then ${}^n g'(t, \cdot, \cdot) \downarrow g'(t, \cdot, \cdot)$ and ${}^n g'(t, \cdot, \cdot)$ satisfies

$$|{}^n g'(t, y_1, z_1) - {}^n g'(t, y_2, z_2)| \leq nu(t)|y_1 - y_2| + nv(t)|z_1 - z_2|, \quad \forall y_1, y_2, z_1, z_2. \quad (15)$$

Furthermore, from the proof of Theorem 1 in Fan, Jiang and Tian [19], we also know that $({}^n y', {}^n z')$, the unique L^2 solution of BSDE($\xi', {}^n g'$), satisfies

$${}^n y' \downarrow y'. \quad (16)$$

Combining (9) and (16) as well as the fact ${}^n g' \geq g'$ we can get that for each $n \geq 1$,

$$\mathbb{1}_{y_t > {}^n y'_t} g(t, y_t, z_t) \leq \mathbb{1}_{y_t > {}^n y'_t} g'(t, y_t, z_t) \leq \mathbb{1}_{y_t > {}^n y'_t} {}^n g'(t, y_t, z_t) \quad \text{dP} \times \text{dt-a.e.}$$

Thus, in view of (15) and the above inequality, Theorem 2.1 yields that

$$\forall n \geq 1, \quad \forall t \in [0, T], \quad y_t \leq {}^n y'_t \quad \text{dP-a.s.} \quad (17)$$

In view of (16), the conclusion follows by letting $n \rightarrow \infty$ in (17). $\square$

By a similar argument to Theorem 2.2 we can get the following Theorem 2.3.

Theorem 2.3. Assume that $0 < T \leq +\infty$, $\xi, \xi' \in L^2(\Omega, \mathcal{F}_T, P)$, g and g' are two generators of BSDEs, and (y', z') is any L^2 solution of BSDE(ξ', g'). Assume further that g satisfies (2A3) and (2A4), and (y, z) is the minimal L^2 solution of BSDE(ξ, g) (recall Lemma 2.2). If $\xi \leq \xi'$, dP-a.s. and (8) holds true, then for each $t \in [0, T]$,

$$y_t \leq y'_t \quad \text{dP-a.s.}$$

Corollary 2.1. Assume that $0 < T \leq +\infty$, $\xi, \xi' \in L^2(\Omega, \mathcal{F}_T, P)$, and both g and g' satisfy (2A3) and (2A4). Let (y, z) and (y', z') be, respectively, the maximal (resp. minimal) L^2 solution of BSDE(ξ, g) and BSDE(ξ', g') (recall Lemma 2.2). If $\xi \leq \xi'$ dP-a.s. and for each $(y, z) \in \mathbb{R} \times \mathbb{R}^d$,

$$g(t, y, z) \leq g'(t, y, z) \quad \text{dP} \times \text{dt-a.e.},$$

then for each $t \in [0, T]$,

$$y_t \leq y'_t \quad dP\text{-a.s.}$$

The following [Theorem 2.4](#) establishes a general comparison theorem for L^1 solutions of BSDEs, which improves considerably several corresponding comparison results obtained respectively in Briand and Hu [6], Fan and Liu [20], Xiao, Li and Fan [39], Fan and Jiang [16] and Tian, Jiang and Shi [38] even for the case of finite time horizon.

Theorem 2.4. Assume that $0 < T \leq +\infty$, $\xi, \xi' \in L^1(\Omega, \mathcal{F}_T, P)$, g and g' are two generators of BSDEs, and (y, z) and (y', z') are, respectively, an L^1 solution of $\text{BSDE}(\xi, g)$ and $\text{BSDE}(\xi', g')$. If $\xi \leq \xi'$ dP -a.s. and one of the following two statements is satisfied:

- (i) g satisfies (2A1), (2A2) and (2A5) (or (2A5')), and (8) holds true;
- (ii) g' satisfies (2A1), (2A2) and (2A5) (or (2A5')), and (9) holds true,

then for each $t \in [0, T]$,

$$y_t \leq y'_t \quad dP\text{-a.s.}$$

Proof. It follows from [Theorem 2.1](#) that we need only to show under the assumptions of [Theorem 2.4](#) that $(y - y')^+$ belongs to $\mathcal{S}$.

We now assume that $\xi \leq \xi'$ dP -a.s., g satisfies (2A1), (2A2) and (2A5), and (8) holds true. The same arguments as follows can prove the other cases. Let us fix $k \in \mathbb{N}^*$ and denote the stopping time

$$\tau_k := \inf \left\{ t \in [0, T] : \int_0^t (|z_s|^2 + |z'_s|^2) \, ds \geq k \right\} \wedge T.$$

Tanaka's formula leads to the equation, setting $\hat{y}_t = y_t - y'_t$, $\hat{z}_t = z_t - z'_t$,

$$\hat{y}_{t \wedge \tau_k}^+ \leq \hat{y}_{\tau_k}^+ + \int_{t \wedge \tau_k}^{\tau_k} \mathbb{1}_{\hat{y}_s > 0} (g(s, y_s, z_s) - g'(s, y'_s, z'_s)) \, ds - \int_{t \wedge \tau_k}^{\tau_k} \mathbb{1}_{\hat{y}_s > 0} \hat{z}_s \cdot dB_s.$$

Since $\mathbb{1}_{\hat{y}_s > 0} (g(s, y'_s, z'_s) - g'(s, y'_s, z'_s))$ is non-positive, we have

$$\begin{aligned} & \mathbb{1}_{\hat{y}_s > 0} (g(s, y_s, z_s) - g'(s, y'_s, z'_s)) \\ &= \mathbb{1}_{\hat{y}_s > 0} (g(s, y_s, z_s) - g(s, y'_s, z'_s)) + \mathbb{1}_{\hat{y}_s > 0} (g(s, y'_s, z'_s) - g'(s, y'_s, z'_s)) \\ &\leq \mathbb{1}_{\hat{y}_s > 0} (g(s, y_s, z_s) - g(s, y'_s, z_s)) + \mathbb{1}_{\hat{y}_s > 0} (g(s, y'_s, z_s) - g(s, y'_s, z'_s)) \end{aligned}$$

and we deduce, using the assumptions (2A1) and (2A5) of g , that

$$\mathbb{1}_{\hat{y}_s > 0} (g(s, y_s, z_s) - g'(s, y'_s, z'_s)) \leq u(s) \rho(\hat{y}_s^+) + 2\lambda(s) (f_s + |y'_s| + |z_s| + |z'_s|)^\alpha.$$

Thus, we get that, with $\phi_s := 2\lambda(s) (f_s + |y'_s| + |z_s| + |z'_s|)^\alpha$,

$$\hat{y}_{t \wedge \tau_k}^+ \leq \hat{y}_{\tau_k}^+ + \int_{t \wedge \tau_k}^{\tau_k} (u(s) \rho(\hat{y}_s^+) + \phi_s) \, ds - \int_{t \wedge \tau_k}^{\tau_k} \mathbb{1}_{\hat{y}_s > 0} \hat{z}_s \cdot dB_s,$$

and then that

$$\hat{y}_{t \wedge \tau_k}^+ \leq \mathbb{E} \left[\hat{y}_{\tau_k}^+ + \int_{t \wedge \tau_k}^{\tau_k} (u(s) \rho(\hat{y}_s^+) + \phi_s) \, ds \middle| \mathcal{F}_t \right]. \quad (18)$$

Furthermore, since $\rho(\cdot)$ is of linear growth, we can find a pair of positive constants k_1 and k_2 such that

$$\rho(u) \leq k_1 + k_2 u, \quad \forall u \geq 0. \quad (19)$$

Then, since both (y, z) and (y', z') are the L^1 solutions, we can send k to ∞ in (18) and use Lebesgue's dominated convergence theorem and Fubini's theorem, in view of $\xi \leq \xi', \tau_k \rightarrow T$ as $k \rightarrow \infty, u(\cdot) \in L^1([0, T]; \mathbb{R}_+)$ and (19), to get that, for each $t \in [0, T]$,

$$\begin{aligned} \hat{y}_t^+ &\leq \mathbb{E} \left[\int_0^T \phi_s \, ds \middle| \mathcal{F}_t \right] + \mathbb{E} \left[\int_t^T u(s) \rho(\hat{y}_s^+) \, ds \middle| \mathcal{F}_t \right] \\ &\leq k_1 \int_0^T u(s) \, ds + \mathbb{E} \left[\int_0^T \phi_s \, ds \middle| \mathcal{F}_t \right] + k_2 \int_t^T u(s) \mathbb{E} [\hat{y}_s^+ | \mathcal{F}_t] \, ds, \end{aligned}$$

and then for each $r \in [t, T]$,

$$\mathbb{E} [\hat{y}_r^+ | \mathcal{F}_t] \leq k_1 \int_0^T u(s) \, ds + \mathbb{E} \left[\int_0^T \phi_s \, ds \middle| \mathcal{F}_t \right] + k_2 \int_r^T u(s) \mathbb{E} [\hat{y}_s^+ | \mathcal{F}_t] \, ds.$$

Gronwall's inequality yields that for each $r \in [t, T]$,

$$\mathbb{E} [\hat{y}_r^+ | \mathcal{F}_t] \leq \left(k_1 \int_0^T u(s) \, ds + \mathbb{E} \left[\int_0^T \phi_s \, ds \middle| \mathcal{F}_t \right] \right) \cdot e^{k_2 \int_r^T u(s) \, ds},$$

from where, by letting $r = t$, we have

$$\hat{y}_t^+ \leq \left(k_1 \int_0^T u(s) \, ds + \mathbb{E} \left[\int_0^T \phi_s \, ds \middle| \mathcal{F}_t \right] \right) \cdot e^{k_2 \int_0^T u(s) \, ds}. \quad (20)$$

Now, let β be any constant which belongs to $(\alpha, 1)$. Then we have

$$\mathbb{E} \left[\left(\int_0^T \phi_s \, ds \right)^{\frac{\beta}{\alpha}} \right] < +\infty. \quad (21)$$

Indeed, Hölder's inequality yields that

$$\begin{aligned} \int_0^T \lambda(s) f_s^\alpha \, ds &\leq \left(\int_0^T \lambda^{\frac{1}{1-\alpha}}(s) \, ds \right)^{1-\alpha} \left(\int_0^T f_s \, ds \right)^\alpha, \\ \int_0^T \lambda(s) |z_s|^\alpha \, ds &\leq \left(\int_0^T \lambda^{\frac{2}{2-\alpha}}(s) \, ds \right)^{\frac{2-\alpha}{2}} \left(\int_0^T |z_s|^2 \, ds \right)^{\frac{\alpha}{2}}, \end{aligned}$$

and z'_s has a similar estimate. Besides,

$$\int_0^T \lambda(s) |y'_s|^\alpha \, ds \leq \int_0^T \lambda(s) \, ds \cdot \sup_{t \in [0, T]} |y'_t|^\alpha.$$

Thus, coming back to the definition of ϕ_s and noticing the assumptions on deterministic function $\lambda(\cdot)$ and the facts that $1 \leq 2/(2-\alpha) \leq 1/(1-\alpha)$ for $\alpha \in (0, 1)$, $f_s \in L^1(\Omega \times [0, T])$,

$(z_t)_{t \in [0, T]}$ and $(z'_t)_{t \in [0, T]}$ belong to the space M^β , and $(y'_t)_{t \in [0, T]}$ belongs to the space $\mathcal{S}^\beta$, we can obtain (21).

Finally, taking supremum over t and expectation on both sides of (20) after taking the power of β/α , and then making use of Doob's inequality, we can get that there exists a constant $\bar{k} > 0$ such that, in view of (21),

$$\mathbb{E} \left[\sup_{t \in [0, T]} |\hat{y}_t^+|^{\frac{\beta}{\alpha}} \right] \leq \bar{k} + \bar{k} \mathbb{E} \left[\left(\int_0^T \phi_s \, ds \right)^{\frac{\beta}{\alpha}} \right] < +\infty,$$

which means that $(y - y')^+ \in \mathcal{S}$. Thus, we complete the proof of Theorem 2.4. $\square$

3. Existence of bounded solutions

In this section, we will put forward and prove an existence result and two comparison results for the maximal and minimal bounded solutions. We denote by $\mathcal{L}(\mathbb{R}; \mathbb{R}^+)$ the set of continuous and strictly positive functions $l(x) : \mathbb{R} \mapsto \mathbb{R}^+$ satisfying

$$\int_{-\infty}^0 \frac{dx}{l(x)} = \int_0^{+\infty} \frac{dx}{l(x)} = +\infty.$$

We will use the following assumptions on the generator g , where $0 < T \leq +\infty$:

(3A1) There exist a function $u(\cdot) \in L^1([0, T]; \mathbb{R}_+)$ and two continuous functions $l(\cdot) \in \mathcal{L}(\mathbb{R}; \mathbb{R}^+)$ and $h(\cdot) : \mathbb{R} \mapsto \mathbb{R}_+$ such that $dP \times dt$ -a.e.,

$$g(\omega, t, y, z) \operatorname{sgn}(y) \leq u(t)l(y) + h(y)|z|^2, \quad \forall y, z.$$

(3A2) There exist a function $\bar{u}(\cdot) \in L^1([0, T]; \mathbb{R}_+)$ and two continuous functions $\bar{\varphi}(\cdot), \bar{h}(\cdot) : \mathbb{R} \mapsto \mathbb{R}_+$ such that $dP \times dt$ -a.e.,

$$|g(\omega, t, y, z)| \leq \bar{u}(t)\bar{\varphi}(y) + \bar{h}(y)|z|^2, \quad \forall y, z.$$

To state the main results of this section, we introduce the following Lemma 3.1.

Lemma 3.1. Assume that $0 < T \leq +\infty$, $u(\cdot) \in L^1([0, T]; \mathbb{R}_+)$ and $l(x) : \mathbb{R} \mapsto \mathbb{R}^+$ is a continuous function. Then $l(\cdot) \in \mathcal{L}(\mathbb{R}; \mathbb{R}^+)$ if and only if for each $-\infty < a \leq 0 \leq b < +\infty$, the following two backward ODEs:

$$L_t = a - \int_t^T u(s) l(L_s) \, ds \tag{22}$$

and

$$U_t = b + \int_t^T u(s) l(U_s) \, ds \tag{23}$$

have both global bounded solutions on $[0, T]$.

Moreover, if $l(\cdot) \in \mathcal{L}(\mathbb{R}; \mathbb{R}^+)$, then (22) and (23) have unique global bounded solutions L_t and U_t on $[0, T]$, and for each $t \in [0, T]$, we have

$$L_0 \leq L_t \leq a \leq 0 \leq b \leq U_t \leq U_0.$$

Proof. In the proof of Lemma 1 in Lepeltier and San Martín [28], by replacing the term $T - t$ with

$$\int_t^T u(s) \, ds$$

we can complete the proof of this lemma. $\square$

The following Theorem 3.1 is one of the main results in this section, which generalizes considerably the corresponding existence results for bounded solutions of BSDEs obtained respectively in Kobylanski [26], Lepeltier and San Martín [28], Briand and Hu [7] and Briand, Lepeltier and San Martín [8] even for the case of finite time horizon.

Theorem 3.1. Assume that $0 < T \leq +\infty$ and that g satisfies (2A3), (3A1) and (3A2). Then for each $\xi \in L^\infty(\Omega, \mathcal{F}_T, P)$, BSDE(ξ, g) has both a minimal and a maximal solution among all bounded solutions (Y, Z) . Moreover, for each $t \in [0, T]$,

$$L_0 \leq L_t \leq Y_t \leq U_t \leq U_0 \quad dP\text{-a.s.},$$

where (L, U) are the unique solutions of (22) and (23) with $a = -\|\xi\|_\infty$ and $b = \|\xi\|_\infty$.

To prove Theorem 3.1, we need the following three lemmas. First, by a similar argument to the proof of Lemma 3 in Lepeltier and San Martín [28] and in view of Lemma 2.2 and Theorem 2.1 (or Theorem 1.2 in Chen and Wang [11]), we can obtain the following Lemma 3.2.

Lemma 3.2. Assume that $0 < T \leq +\infty$, $u(\cdot) \in L^1([0, T]; \mathbb{R}_+)$, $v(\cdot) \in L^2([0, T]; \mathbb{R}_+)$, $f_1(\cdot) : \mathbb{R} \mapsto \mathbb{R}$ is a continuous function with linear growth and the random function $f_2(\omega, t, z) : \Omega \times [0, T] \times \mathbb{R}^d \mapsto \mathbb{R}$ is $\mathcal{P} \times \mathcal{B}(\mathbb{R}^d)$ -measurable, with $dP \times dt$ -a.e.,

$$f_2(\omega, t, 0) = 0 \quad \text{and} \quad |f_2(\omega, t, z_1) - f_2(\omega, t, z_2)| \leq v(t)|z_1 - z_2|, \quad \forall z_1, z_2 \in \mathbb{R}^d.$$

If the backward ODE

$$J_t = a + \int_t^T u(s) f_1(J_s) \, ds, \quad a \in \mathbb{R}$$

has a unique solution J on $[0, T]$, then the BSDE

$$Y_t = a + \int_t^T (u(s) f_1(Y_s) + f_2(s, Z_s)) \, ds - \int_t^T Z_s \cdot dB_s, \quad t \in [0, T]$$

has a unique solution given by $Z \equiv 0$ and $Y \equiv J$.

The following Lemma 3.3 is the first step to prove Theorem 3.1.

Lemma 3.3. Assume that $0 < T \leq +\infty$ and $\eta \in L^\infty(\Omega, \mathcal{F}_T, P)$ which satisfies $0 \leq \alpha \leq \eta \leq \beta$ dP -a.s. We assume without loss of generality that $\alpha \leq 1$ and $\beta \geq 1$. Suppose that $G(\omega, t, y, z) : \Omega \times [0, T] \times \mathbb{R} \times \mathbb{R}^d \mapsto \mathbb{R}$ vanishes when $y \leq 0$ and verifies for a constant $k > 0$ and a function $u(\cdot) \in L^1([0, T]; \mathbb{R}_+)$ the following restriction:

$$dP \times dt\text{-a.e.}, \quad \forall y > 0, \quad \forall z \in \mathbb{R}^d, \quad -u(t)y - k|z|^2 \leq G(\omega, t, y, z) \leq u(t)y.$$

Also we assume that G is $\mathcal{P} \otimes \mathcal{B}(\mathbb{R}^{d+1})$ measurable and $dP \times dt$ -a.e., $G(\omega, t, \cdot, \cdot)$ is continuous. Then the BSDE

$$Y_t = \eta + \int_t^T G(s, Y_s, Z_s) \, ds - \int_t^T Z_s \cdot dB_s, \quad t \in [0, T]$$

has a maximal bounded solution (θ, Γ) . Moreover, for each $t \in [0, T]$, we have $Q_0 \leq Q_t \leq \theta_t \leq S_t \leq S_0$ dP -a.s., where

$$Q_t := \alpha \exp\left(-\int_t^T u(s) ds\right), \quad S_t := \beta \exp\left(\int_t^T u(s) ds\right).$$

Proof. We will follow those steps used in the proof of Theorem 2 in Lepeltier and San Martín [28]. We take $\kappa_n : \mathbb{R}^d \mapsto \mathbb{R}$ a sequence of smooth functions such that

$$(1) 0 \leq \kappa_n \leq 1; \quad (2) \kappa_n(z) = 1 \quad \text{if } |z| \leq n; \quad (3) \kappa_n(z) = 0 \quad \text{if } |z| \geq n+1.$$

For each $(\omega, t, y, z) \in \Omega \times [0, T] \times \mathbb{R} \times \mathbb{R}^d$ and each $n \geq 1$, define

$$G_n(\omega, t, y, z) := u(t)y\mathbb{1}_{y>0}\left(1 - \kappa_n\left(z \cdot e^{\frac{t}{2}}\right)\right) + \kappa_n\left(z \cdot e^{\frac{t}{2}}\right)G(\omega, t, y, z).$$

It is immediately seen that $G_n \downarrow G$ and that G_n is a continuous function of (y, z) which satisfies that $dP \times dt$ -a.e.,

$$|G_n(\omega, t, y, z)| \leq k(n+1)^2 e^{-t} + u(t)|y|, \quad \forall y, z. \quad (24)$$

Then by Lemma 2.2 we have the existence of a maximal L^2 solution (θ^n, Γ^n) for the equation

$$\theta_t^n = \eta + \int_t^T G_n(s, \theta_s^n, \Gamma_s^n) ds - \int_t^T \Gamma_s^n \cdot dB_s.$$

Since $dP \times dt$ -a.e.,

$$\lambda_n(t, y, z) := -u(t)|y| - k|z|^2 \kappa_n\left(z \cdot e^{\frac{t}{2}}\right) \leq G_n(\omega, t, y, z) \leq u(t)|y|, \quad \forall y, z,$$

noticing that both $u(t)|y|$ and λ_n satisfy (2A1)–(2A4), from Theorem 2.1 we can get that for each $t \in [0, T]$,

$$Q_t^n \leq \theta_t^n \leq \bar{S}_t \quad dP\text{-a.s.},$$

where (Q^n, Z^n) and $(\bar{S}, \bar{Z})$ are, respectively, the unique L^2 solution of BSDE (α, λ_n) and BSDE $(\beta, u(t)|y|)$ by Lemma 2.2 and Theorem 2.1. Furthermore, it follows from Lemma 3.2 that $(\bar{S}, \bar{Z}) = (S, 0)$ and for each $n \geq 1$, $(Q^n, Z^n) = (Q, 0)$.

Consequently, θ^n is a decreasing and bounded sequence (in view of (24) and Corollary 2.1), then we have the existence of θ such that for each $t \in [0, T]$, $\theta_t^n \downarrow \theta_t$ dP -a.s. and

$$\mathbb{E}\left[\int_0^T u(t)|\theta_t^n - \theta_t|^2 dt\right] \rightarrow 0, \quad \text{as } n \rightarrow \infty.$$

Moreover θ satisfies that for each $t \in [0, T]$, $Q_0 \leq Q_t \leq \theta_t \leq S_t \leq S_0$ dP -a.s.

In the sequel, following closely the proof procedure of Theorem 2 in Lepeltier and San Martín [28] and noticing that $dP \times dt$ -a.e., for each $(y, z) \in [Q_0, S_0] \times \mathbb{R}^d$,

$$\sup_{n \geq 1} |G_n(\omega, t, y, z)| \leq u(t)S_0 + k|z|^2,$$

we can prove that Γ^n has a convergent subsequence in M^2 . Thus, take Γ any accumulation point in M^2 of Γ^n then it is easy to verify that (θ, Γ) is a solution of BSDE (η, G) (for more details see the proof of Theorem 1 in Fan, Jiang and Tian [19]).

Finally, for any bounded solution $(\hat{Y}, \hat{Z})$ of BSDE (η, G) , noticing that $G_n \downarrow G$, (24) and the fact that (θ^n, I^n) is the maximal L^2 solution of BSDE (η, G^n) , by Theorem 2.2 we can conclude that for each $t \in [0, T]$ and each $n \geq 1$,

$$\hat{Y}_t \leq \theta_t^n \quad dP\text{-a.s.},$$

and then $\hat{Y} \leq \theta$. Thus, Lemma 3.3 is proved. $\square$

By virtue of Lemma 3.3 we can prove the following Lemma 3.4.

Lemma 3.4. Assume that $0 < T \leq +\infty$ and the generator g satisfies (2A3). Assume further that there exist a constant $\gamma > 0$ and a function $u(\cdot) \in L^1([0, T]; \mathbb{R}_+)$ such that $dP \times dt$ -a.e.,

$$|g(\omega, t, y, z)| \leq u(t) + \frac{\gamma}{2}|z|^2, \quad \forall y, z. \quad (25)$$

Then for each $\xi \in L^\infty(\Omega, \mathcal{F}_T, P)$, BSDE (ξ, g) has both a maximal and a minimal solution among all bounded solutions (y, z) . Moreover, for each $t \in [0, T]$, we have

$$|y_t| \leq \|\xi\|_\infty + \int_t^T u(s) ds \quad dP\text{-a.s.}$$

Proof. We will first prove the existence of the maximal bounded solution by following those steps used in the proof of Theorem 1 in Lepeltier and San Martín [28]. Let

$$\eta := e^{\gamma\xi} \in L^\infty(\Omega, \mathcal{F}_T, P), \quad \alpha := e^{-\gamma\|\xi\|_\infty}, \quad \beta := e^{\gamma\|\xi\|_\infty},$$

$$Q_t = \alpha \exp\left(-\gamma \int_t^T u(s) ds\right), \quad S_t = \beta \exp\left(\gamma \int_t^T u(s) ds\right), \quad t \in [0, T],$$

and for each $(\omega, t, y, z) \in \Omega \times [0, T] \times \mathbb{R} \times \mathbb{R}^d$, define

$$G(\omega, t, y, z) := \mathbb{1}_{y>0} \left(\gamma y g\left(\omega, t, \frac{\ln y}{\gamma}, \frac{z}{\gamma y}\right) - \frac{1}{2} \frac{|z|^2}{y} \right).$$

It then follows from (25) that $0 < \alpha \leq 1 \leq \beta$, $\alpha \leq \eta \leq \beta$ dP -a.s., and $dP \times dt$ -a.e.,

$$\forall y > 0, \forall z \in \mathbb{R}^d, \quad -\gamma u(t)y - \frac{|z|^2}{y} \leq G(\omega, t, y, z) \leq \gamma u(t)y.$$

Furthermore, for each pair of positive real numbers K_1 and K_2 satisfying that $[Q_0, S_0] \subset [2K_1, K_2/2]$, let Ψ be a smooth function with values in $[0, 1]$ which satisfies that $\Psi(x) = 1$ for $x \in [2K_1, K_2/2]$, and $\Psi(x) = 0$ when x is outside $[K_1, K_2]$. Define

$$G_\Psi(\omega, t, y, z) := \Psi(y)G(\omega, t, y, z),$$

then $dP \times dt$ -a.e.,

$$\forall y > 0, \forall z \in \mathbb{R}^d, \quad -\gamma u(t)y - \left(\frac{1}{K_1}\right)|z|^2 \leq G_\Psi(\omega, t, y, z) \leq \gamma u(t)y.$$

Thanks to Lemma 3.3, we know that the BSDE

$$Y_t = \eta + \int_t^T G_\Psi(s, Y_s, Z_s) ds - \int_t^T Z_s \cdot dB_s, \quad t \in [0, T]$$

has a maximal one (Y^Ψ, Z^Ψ) among all bounded solutions. Moreover, we also have that for each $t \in [0, T]$, $0 < Q_0 \leq Q_t \leq Y_t^\Psi \leq S_t \leq S_0$ dP -a.s., which means that, in view of the definition of Ψ , (Y^Ψ, Z^Ψ) is a bounded solution to the following BSDE:

$$Y_t = \eta + \int_t^T G(s, Y_s, Z_s) \, ds - \int_t^T Z_s \cdot dB_s, \quad t \in [0, T].$$

We define

$$y^\Psi := \frac{\ln(Y^\Psi)}{\gamma} \quad \text{and} \quad z^\Psi := \frac{Z^\Psi}{\gamma Y^\Psi}.$$

It follows from Itô's formula that (y^Ψ, z^Ψ) is a bounded solution of $\text{BSDE}(\xi, g)$, and it is easy to verify that for each $t \in [0, T]$,

$$|y_t^\Psi| \leq \|\xi\|_\infty + \int_t^T u(s) \, ds \quad dP\text{-a.s.} \quad (26)$$

In the sequel, let us show that (y^Ψ, z^Ψ) is also the maximal bounded solution. Indeed, let $(\hat{y}, \hat{z})$ be a bounded solution of $\text{BSDE}(\xi, g)$, with $A \leq \hat{y} \leq B$. We can choose positive real numbers $\hat{K}_1, \hat{K}_2$ satisfying that $[e^{\gamma A}, e^{\gamma B}] \subset [2\hat{K}_1, \hat{K}_2]$ and consider $\hat{\Psi}$ with values in $[0, 1]$ which satisfies that $\hat{\Psi}(x) = 1$ for $x \in [2\hat{K}_1, \hat{K}_2/2]$, and $\hat{\Psi}(x) = 0$ when x is outside $[\hat{K}_1, \hat{K}_2]$. It then follows from Itô's formula that $\hat{Y} := e^{\gamma \hat{y}}, \hat{Z} := \gamma \hat{z} \hat{y}$ is a bounded solution of $\text{BSDE}(\eta, G_{\hat{\Psi}})$, where $G_{\hat{\Psi}}(\omega, t, y, z) := \hat{\Psi}(y)G(\omega, t, y, z)$. Note that $dP \times dt$ -a.e.,

$$\forall y > 0, \forall z \in \mathbb{R}^d, \quad -\gamma u(t)y - \left(\frac{1}{\hat{K}_1}\right)|z|^2 \leq G_{\hat{\Psi}}(\omega, t, y, z) \leq \gamma u(t)y.$$

Thanks to [Lemma 3.3](#) again, we know that $0 < Q_0 \leq Q_t \leq \hat{Y}_t \leq S_t \leq S_0$, which means that $(\hat{Y}, \hat{Z})$ is also a bounded solution of $\text{BSDE}(\eta, G_\Psi)$. Therefore, since (Y^Ψ, Z^Ψ) is the maximal bounded solution of $\text{BSDE}(\eta, G_\Psi)$, we have $e^{\gamma \hat{y}} = \hat{Y} \leq Y^\Psi = e^{\gamma y^\Psi}$, and then for each $t \in [0, T]$, $\hat{y}_t \leq y_t^\Psi$ dP -a.s. This is the desired result. This argument also shows that y^Ψ does not depend on Ψ .

Finally, let us define

$$\tilde{g}(\omega, t, y, z) := -g(\omega, t, -y, -z), \quad \forall \omega, t, y, z.$$

Then $\tilde{g}$ also satisfies (2A3) and (25). Consequently, by above arguments we know that $\text{BSDE}(-\xi, \tilde{g})$ has a maximal bounded solution $(\tilde{y}, \tilde{z})$ and $\tilde{y}$ also satisfies the estimate in (26). Furthermore, it is easy to verify that $(-\tilde{y}, -\tilde{z})$ is just the minimal bounded solution of $\text{BSDE}(\xi, g)$. [Lemma 3.4](#) is then proved. $\square$

Remark 3.1. A similar result to [Lemma 3.4](#) was given in Morlais [30] with a different method.

We are now in a position to prove [Theorem 3.1](#).

Proof of Theorem 3.1. Assume that $\xi \in L^\infty(\Omega, \mathcal{F}_T, P)$ and g satisfies (2A3), (3A1) and (3A2). We only prove the case of maximal solution, the other case can be proved in a similar way. First, by [Lemma 3.1](#) we can let $(L_t)_{t \in [0, T]}$ and $(U_t)_{t \in [0, T]}$ be, respectively, the unique global solution to the following two backward ODEs

$$L_t = -\|\xi\|_\infty - \int_t^T u(s) l(L_s) \, ds \quad (27)$$

and

$$U_t = \|\xi\|_\infty + \int_t^T u(s) l(U_s) \, ds. \quad (28)$$

Then for each $t \in [0, T]$, we have

$$L_0 \leq L_t \leq -\|\xi\|_\infty \leq 0 \leq \|\xi\|_\infty \leq U_t \leq U_0.$$

For each constant $K > 0$ satisfying that $[L_0, U_0] \subset [-K, K]$, consider a continuous function κ such that $\kappa(x) = -K$ when $x < -K$, $\kappa(x) = x$ when $x \in [-K, K]$, and $\kappa(x) = K$ when $x > K$. For each $(\omega, t, y, z) \in \Omega \times [0, T] \times \mathbb{R} \times \mathbb{R}^d$, we define

$$g_\kappa(\omega, t, y, z) := g(\omega, t, \kappa(y), z) \quad \text{and} \quad \gamma^K := 2 \left(\max_{x \in [-K, K]} h(x) + 1 \right).$$

Then it follows from (3A1) and (3A2) that $dP \times dt$ -a.e., for each $(y, z) \in \mathbb{R} \times \mathbb{R}^d$,

$$g_\kappa(t, y, z) \operatorname{sgn}(y) \leq u(t)l(\kappa(y)) + \frac{\gamma^K}{2} |z|^2 \quad (29)$$

and

$$|g_\kappa(t, y, z)| \leq \bar{u}(t) \left(\max_{x \in [-K, K]} \bar{\varphi}(x) \right) + \left(\max_{x \in [-K, K]} \bar{h}(x) \right) |z|^2.$$

It then follows from [Lemma 3.4](#) that BSDE(ξ, g_κ) has a maximal bounded solution $(y_t^\kappa, z_t^\kappa)_{t \in [0, T]}$. Let $Y_t^\kappa := e^{\gamma^K y_t^\kappa}$ and $Z_t^\kappa := \gamma^K Y_t^\kappa z_t^\kappa$, then (Y^κ, Z^κ) is a bounded solution to the following BSDE

$$Y_t = \eta + \int_t^T G_\kappa(s, Y_s, Z_s) \, ds - \int_t^T Z_s \cdot dB_s, \quad t \in [0, T],$$

where $\eta := e^{\gamma^K \xi}$ and for each $(\omega, t, y, z) \in \Omega \times [0, T] \times \mathbb{R} \times \mathbb{R}^d$,

$$G_\kappa(\omega, t, y, z) := \left[\gamma^K y g_\kappa \left(\omega, t, \frac{\ln y}{\gamma^K}, \frac{z}{\gamma^K y} \right) - \frac{1}{2} \frac{|z|^2}{y} \right] \mathbb{1}_{y>0}.$$

Furthermore, it follows from (29) that $dP \times dt$ -a.e.,

$$\forall y > 0, \forall z \in \mathbb{R}^d, \quad g_\kappa(t, y, z) \leq u(t)l(\kappa(y)) + \frac{\gamma^K}{2} |z|^2, \quad (30)$$

and then $dP \times dt$ -a.e.,

$$\forall y > 1, \forall z \in \mathbb{R}^d, \quad G_\kappa(\omega, t, y, z) \leq G'_\kappa(\omega, t, y, z) := \gamma^K y u(t)l \left(\kappa \left(\frac{\ln y}{\gamma^K} \right) \right) \mathbb{1}_{y>0}. \quad (31)$$

Note that $dP \times dt$ -a.e., for each $(y, z) \in \mathbb{R} \times \mathbb{R}^d$,

$$|G'_\kappa(t, y, z)| \leq \gamma^K \left(\max_{x \in [-K, K]} l(x) \right) u(t)|y|. \quad (32)$$

By [Lemma 2.2](#) we know that BSDE($\|\eta\|_\infty, G'_\kappa$) has a maximal L^2 solution $({}^\kappa Y', {}^\kappa Z')$. On the other hand, noticing the definition of κ and the assumptions of $u(\cdot)$ and $l(\cdot)$, we can verify directly

that the following backward ODE

$$R_t^K = e^{\gamma^K \|\xi\|_\infty} + \int_t^T \gamma^K u(s) l \left(\kappa \left(\frac{\ln R_s^K}{\gamma^K} \right) \right) R_s^K \mathbb{1}_{R_s^K > 0} \, ds$$

has a unique solution $R_t^K = e^{\gamma^K U_t}$ with $t \in [0, T]$, where U_t is defined in (28). Thus, by terms of Lemma 3.2 we know that $({}^\kappa Y', {}^\kappa Z') = (R^K, 0)$. Furthermore, in view of (31), (32) and the fact ${}^\kappa Y' \geq 1$, it follows from Theorem 2.2 and Remark 2.2 that for each $t \in [0, T]$,

$$e^{\gamma^K y_t^K} = Y_t^K \leq {}^\kappa Y_t' = e^{\gamma^K U_t} \quad dP\text{-a.s.},$$

and then

$$y_t^K \leq U_t \leq U_0 \leq K \quad dP\text{-a.s.} \quad (33)$$

On the other hand, noticing that $\kappa(-x) = -\kappa(x)$, by (29) we get that $dP \times dt$ -a.e.,

$$\forall y > 0, \forall z \in \mathbb{R}^d, \quad \bar{g}_\kappa(\omega, t, y, z) := -g_\kappa(\omega, t, -y, -z) \leq u(t) \bar{l}(\kappa(y)) + \frac{\gamma^K}{2} |z|^2,$$

where $\bar{l}(u) := l(-u)$ for each $u \in \mathbb{R}$. Furthermore, note that $(-y_t^K, -z_t^K)_{t \in [0, T]}$ is a bounded solution of BSDE $(-\xi, \bar{g}_\kappa)$, $\bar{l}(\cdot) \in \mathcal{L}(\mathbb{R}; \mathbb{R}^+)$ and the following backward ODE

$$\bar{U}_t = \|\xi\|_\infty + \int_t^T u(s) \bar{l}(\bar{U}_s) \, ds = \|\xi\|_\infty + \int_t^T u(s) l(-\bar{U}_s) \, ds$$

has a unique solution $\bar{U}_t = -L_t$ with $t \in [0, T]$, where L_t is defined in (27). The similar argument to that from (29) to (33) yields that for each $t \in [0, T]$,

$$-y_t^K \leq \bar{U}_t = -L_t \quad dP\text{-a.s.},$$

and then

$$y_t^K \geq L_t \geq L_0 \geq -K \quad dP\text{-a.s.} \quad (34)$$

Thus, in view of (33), (34), the definition of g_κ and κ , we know that the $(y_t^K, z_t^K)_{t \in [0, T]}$ is a bounded solution of BSDE (ξ, g) .

Finally, let us show that $(y_t^K, z_t^K)_{t \in [0, T]}$ is also the maximal bounded solution of BSDE (ξ, g) . Indeed, let $(\hat{y}_t, \hat{z}_t)_{t \in [0, T]}$ be a bounded solution of BSDE (ξ, g) , with $A \leq \hat{y} \leq B$. We choose a positive constant $\hat{K}$ satisfying that $[A, B] \cup [L_0, U_0] \subset [-\hat{K}, \hat{K}]$ and consider $\hat{\kappa}$ such that $\hat{\kappa}(x) = -\hat{K}$ for $x < -\hat{K}$, $\hat{\kappa}(x) = x$ for $x \in [-\hat{K}, \hat{K}]$, and $\hat{\kappa}(x) = \hat{K}$ for $x > \hat{K}$. For each $(\omega, t, y, z) \in \Omega \times [0, T] \times \mathbb{R} \times \mathbb{R}^d$, we define

$$g_{\hat{\kappa}}(\omega, t, y, z) := g(\omega, t, \hat{\kappa}(y), z).$$

Then $(\hat{y}_t, \hat{z}_t)_{t \in [0, T]}$ is a bounded solution of BSDE $(\xi, g_{\hat{\kappa}})$. Furthermore, by the above argument as above (from (29) to (34)) we can conclude that for each $t \in [0, T]$,

$$-K \leq L_0 \leq L_t \leq \hat{y}_t \leq U_t \leq U_0 \leq K \quad dP\text{-a.s.},$$

which means that $(\hat{y}_t, \hat{z}_t)_{t \in [0, T]}$ is also a bounded solution of BSDE (ξ, g_κ) . Note that $(y_t^K, z_t^K)_{t \in [0, T]}$ is the maximal bounded solution of BSDE (ξ, g_κ) . We know that for each $t \in [0, T]$, $\hat{y}_t \leq y_t^K$ dP -a.s., which is the desired result. This argument also shows that y_t^K does not depend on κ . The proof of Theorem 3.1 is then complete. $\square$

Finally, in view of [Theorems 2.2](#) and [2.3](#), by checking carefully the proof of [Theorem 3.1](#), [Lemma 3.4](#) and, especially, [Lemma 3.3](#), we can prove that the following two comparison theorems for the maximal and minimal bounded solutions hold true.

Theorem 3.2. Assume that $0 < T \leq +\infty$, $\xi, \xi' \in L^\infty(\Omega, \mathcal{F}_T, P)$, g and g' are two generators of BSDEs, and (y, z) is any bounded solution of $\text{BSDE}(\xi, g)$. Assume further that g' satisfies (2A3), (3A1) and (3A2), and (y', z') is the maximal bounded solution of $\text{BSDE}(\xi', g')$ (recall [Theorem 3.1](#)). If $\xi \leq \xi'$ dP -a.s. and (9) holds true, then for each $t \in [0, T]$,

$$y_t \leq y'_t \quad dP\text{-a.s.}$$

Theorem 3.3. Assume that $0 < T \leq +\infty$, $\xi, \xi' \in L^\infty(\Omega, \mathcal{F}_T, P)$, g and g' are two generators of BSDEs, and (y', z') is any bounded solution of $\text{BSDE}(\xi', g')$. Assume further that g satisfies (2A3), (3A1) and (3A2), and (y, z) is the minimal bounded solution of $\text{BSDE}(\xi, g)$ (recall [Theorem 3.1](#)). If $\xi \leq \xi'$ dP -a.s. and (8) holds true, then for each $t \in [0, T]$,

$$y_t \leq y'_t \quad dP\text{-a.s.}$$

Corollary 3.1. Assume that $0 < T \leq +\infty$, $\xi, \xi' \in L^\infty(\Omega, \mathcal{F}_T, P)$, and both g and g' satisfy (2A3), (3A1) and (3A2). Let (y, z) and (y', z') be, respectively, the maximal (resp. minimal) bounded solution of $\text{BSDE}(\xi, g)$ and $\text{BSDE}(\xi', g')$ (recall [Theorem 3.1](#)). If $\xi \leq \xi'$ dP -a.s. and for each $(y, z) \in \mathbb{R} \times \mathbb{R}^d$,

$$g(t, y, z) \leq g'(t, y, z) \quad dP \times dt\text{-a.e.},$$

then for each $t \in [0, T]$, we have

$$y_t \leq y'_t \quad dP\text{-a.s.}$$

4. Comparison theorems of bounded solutions

In this section, we will establish two comparison theorems and an existence and uniqueness theorem for bounded solutions of BSDEs. Let us first introduce the following assumptions on the generator g , where $0 < T \leq +\infty$:

(4A1) There exists a function $v(\cdot) \in L^2([0, T]; \mathbb{R}_+)$ such that $dP \times dt$ -a.e.,

$$|g(\omega, t, y, z_1) - g(\omega, t, y, z_2)| \leq (v(t) + |z_1| + |z_2|)|z_1 - z_2|, \quad \forall y, z_1, z_2.$$

(4A2) $dP \times dt$ -a.e., $\forall y$, $g(\omega, t, y, \cdot) : \mathbb{R}^d \mapsto \mathbb{R}$ is convex, or $dP \times dt$ -a.e., $\forall y$, $g(\omega, t, y, \cdot) : \mathbb{R}^d \mapsto \mathbb{R}$ is concave.

Let us now recall several facts on the martingales of bounded mean oscillation, briefly called BMO-martingales. Readers are referred to Kazamaki [25], Hu, Imkeller and Müller [22] and Briand and Confortola [4] for more details. Suppose that $(z_t)_{t \in [0, T]}$ is an $(\mathcal{F}_t)$ -progressively measurable $\mathbb{R}^d$ -valued process such that $\int_0^T |z_s|^2 ds < +\infty$, dP -a.s. First, it is well known that the local martingale $\int_0^\cdot z_s \cdot dB_s$ is a BMO-martingale if and only if

$$\sup_{\tau \in \Sigma_T} \left| \mathbb{E} \left[\int_\tau^T |z_s|^2 ds \middle| \mathcal{F}_\tau \right] \right|_\infty < +\infty,$$

where and hereafter $|\cdot|_\infty$ denotes the essential supremum norm. Furthermore, according to Theorem 2.3 in Kazamaki [25], the stochastic exponential $\mathcal{E}(M)$ of a BMO-martingale M is a

uniformly integrable martingale, note that the stochastic exponential $\mathcal{E}(M)$ is given by

$$\mathcal{E}(M)_t = \exp\left(M_t - \frac{1}{2}\langle M \rangle_t\right), \quad t \in [0, T],$$

where the quadratic variation is denoted by $\langle M \rangle$. Finally, by Theorem 3.6 in Kazamaki [25] we also know that if Q is a probability measure defined by

$$dQ = \mathcal{E}(M)_T dP$$

for a BMO-martingale M under P , then the Girsanov transform of a BMO-martingale under P is a BMO-martingale under Q .

The following Lemma 4.1 will be used several times later.

Lemma 4.1. *Let $0 < T \leq +\infty$, g satisfies (3A2), and let (y, z) be a bounded solution of BSDE(ξ, g). Then $\int_0^\cdot z_s \cdot dB_s$ is a BMO-martingale under P .*

Proof. Assume that $A \leq y \leq B$ with $A, B \in \mathbb{R}$. Let $\gamma = 2(\max_{x \in [A, B]} \bar{h}(x) + 1)$ and consider the following function from $\mathbb{R}_+$ into itself defined by

$$f(x) = \frac{1}{\gamma^2}(e^{\gamma x} - 1 - \gamma x).$$

Clearly, $x \mapsto f(|x|)$ is $\mathcal{C}^2$ and for each $\tau \in \Sigma_T$, we have from Itô's formula,

$$\begin{aligned} f(|y_\tau|) &= f(|\xi|) + \int_\tau^T \left(f'(|y_s|) \operatorname{sgn}(y_s) g(s, y_s, z_s) - \frac{1}{2} f''(|y_s|) |z_s|^2 \right) ds \\ &\quad - \int_\tau^T f'(|y_s|) \operatorname{sgn}(y_s) z_s \cdot dB_s. \end{aligned}$$

Since $A \leq y \leq B$ and $f'(x) \geq 0$ for $x \geq 0$, by (3A2) and the definition of γ we can get the existence of a constant $k > 0$ satisfying that

$$\begin{aligned} 0 \leq f(|y_\tau|) &\leq k \left(1 + \int_0^T \bar{u}(s) ds \right) - \int_\tau^T f'(|y_s|) \operatorname{sgn}(y_s) z_s \cdot dB_s \\ &\quad - \frac{1}{2} \int_\tau^T \left[(f''(|y_s|) - \gamma f'(|y_s|)) |z_s|^2 \right] ds. \end{aligned}$$

Note that (y, z) be a bounded solution and $f''(x) - \gamma f'(x) = 1$ for $x \geq 0$. By taking the conditional expectation with respect to $\mathcal{F}_\tau$ under P in the previous inequality we get that for each $\tau \in \Sigma_T$,

$$\mathbb{E} \left[\int_\tau^T |z_s|^2 ds \middle| \mathcal{F}_\tau \right] \leq 2k \left(1 + \int_0^T \bar{u}(s) ds \right).$$

That is to say, $\int_0^\cdot z_s \cdot dB_s$ is a BMO-martingale under P . The proof is complete. $\square$

The following Theorems 4.1–4.2 establish two comparison theorems for bounded solutions of BSDEs, which considerably improves the corresponding comparison results obtained in Briand and Hu [7] and Morlais [30] even for the case of finite time horizon.

Theorem 4.1. *Assume that $0 < T \leq +\infty$, $\xi, \xi' \in L^\infty(\Omega, \mathcal{F}_T, P)$, both g and g' satisfy (3A2), and (y, z) and (y', z') are, respectively, a bounded solution of BSDE(ξ, g) and BSDE(ξ', g'). If $\xi \leq \xi'$ dP -a.s. and one of the following two statements is satisfied:*

- (i) g satisfies (2A1) and (4A1), and (8) holds true;
(ii) g' satisfies (2A1) and (4A1), and (9) holds true,

then for each $t \in [0, T]$,

$$y_t \leq y'_t \quad dP\text{-a.s.}$$

Proof. We only prove the case (i). The other case can be proved in the same way. Tanaka's formula leads to the equation, setting $\hat{y}_t = y_t - y'_t$, $\hat{z}_t = z_t - z'_t$ and noticing that dP -a.s., $(\xi - \xi')^+ = 0$,

$$\hat{y}_t^+ \leq \int_t^T \mathbb{1}_{\hat{y}_s > 0} (g(s, y_s, z_s) - g'(s, y'_s, z'_s)) ds - \int_t^T \mathbb{1}_{\hat{y}_s > 0} \hat{z}_s \cdot dB_s, \quad t \in [0, T]. \quad (35)$$

First of all, since $\mathbb{1}_{\hat{y}_s > 0} (g(s, y'_s, z'_s) - g'(s, y'_s, z'_s))$ is non-positive, we have

$$\begin{aligned} & \mathbb{1}_{\hat{y}_s > 0} (g(s, y_s, z_s) - g'(s, y'_s, z'_s)) \\ &= \mathbb{1}_{\hat{y}_s > 0} (g(s, y_s, z_s) - g(s, y'_s, z'_s)) + \mathbb{1}_{\hat{y}_s > 0} (g(s, y'_s, z'_s) - g'(s, y'_s, z'_s)) \\ &\leq \mathbb{1}_{\hat{y}_s > 0} (g(s, y_s, z_s) - g(s, y'_s, z'_s)) + \mathbb{1}_{\hat{y}_s > 0} (g(s, y'_s, z'_s) - g'(s, y'_s, z'_s)) \end{aligned}$$

and we deduce, using the assumptions (2A1) and (4A1) for g , that

$$\mathbb{1}_{\hat{y}_s > 0} (g(s, y_s, z_s) - g'(s, y'_s, z'_s)) \leq u(s) \rho(\hat{y}_s^+) + \mathbb{1}_{\hat{y}_s > 0} (v(s) + |z_s| + |z'_s|) |\hat{z}_s|. \quad (36)$$

Thus, by (35) and (36) we can get that for each $t \in [0, T]$,

$$\begin{aligned} \hat{y}_t^+ &\leq \int_t^T [u(s) \rho(\hat{y}_s^+) + \mathbb{1}_{\hat{y}_s > 0} (v(s) + |z_s| + |z'_s|) |\hat{z}_s|] ds - \int_t^T \mathbb{1}_{\hat{y}_s > 0} \hat{z}_s \cdot dB_s \\ &= \int_t^T u(s) \rho(\hat{y}_s^+) ds - \int_t^T \mathbb{1}_{\hat{y}_s > 0} \hat{z}_s \cdot \left[dB_s - \frac{(v(s) + |z_s| + |z'_s|) \hat{z}_s}{|\hat{z}_s|} \mathbb{1}_{|\hat{z}_s| \neq 0} ds \right]. \quad (37) \end{aligned}$$

Furthermore, since both g and g' satisfy (3A2), and both (y, z) and (y', z') are bounded solutions, it follows from Lemma 4.1 that both $\int_0^\cdot z_s \cdot dB_s$ and $\int_0^\cdot z'_s \cdot dB_s$ are BMO-martingale under P . Then we have

$$\begin{aligned} & \sup_{\tau \in \Sigma_T} \left| \mathbb{E} \left[\int_\tau^T (v(s) + |z_s| + |z'_s|)^2 ds \middle| \mathcal{F}_\tau \right] \right|_\infty \\ &\leq 4 \int_0^T v^2(s) ds + 4 \sup_{\tau \in \Sigma_T} \left| \mathbb{E} \left[\int_\tau^T |z_s|^2 ds \middle| \mathcal{F}_\tau \right] \right|_\infty \\ &\quad + 4 \sup_{\tau \in \Sigma_T} \left| \mathbb{E} \left[\int_\tau^T |z'_s|^2 ds \middle| \mathcal{F}_\tau \right] \right|_\infty < +\infty, \end{aligned}$$

which means that the process

$$M_t := \int_0^t \frac{(v(s) + |z_s| + |z'_s|) \hat{z}_s}{|\hat{z}_s|} \mathbb{1}_{|\hat{z}_s| \neq 0} \cdot dB_s, \quad t \in [0, T]$$

is a BMO-martingale under P . Then the stochastic exponential

$$\mathcal{E}(M)_t = \exp \left(M_t - \frac{1}{2} \langle M \rangle_t \right), \quad t \in [0, T]$$

of M is a uniformly integrable martingale. Now, let us define by Q the probability measure on $(\Omega, \mathcal{F}_T)$ given by

$$\frac{dQ}{dP} := \mathcal{E}(M)_T.$$

Then, noticing that

$$\overline{M}_t := \int_0^t \mathbb{1}_{\hat{y}_s > 0} \hat{z}_s \cdot dB_s, \quad t \in [0, T]$$

is also a BMO-martingale under P , we know that the process

$$\int_0^t \mathbb{1}_{\hat{y}_s > 0} \hat{z}_s \cdot [dB_s - \frac{(v(s) + |z_s| + |z'_s|)\hat{z}_s}{|\hat{z}_s|} \mathbb{1}_{|\hat{z}_s| \neq 0} ds], \quad t \in [0, T],$$

the Girsanov's transform of $\overline{M}$, is a BMO-martingale under Q . Let $\mathbb{E}_Q[X|\mathcal{F}_t]$ represent the conditional expectation of the random variable X with respect to $\mathcal{F}_t$ under Q . Taking the conditional expectation with respect to $\mathcal{F}_t$ under Q in (37) yields that for each $t \in [0, T]$,

$$\hat{y}_t^+ \leq \mathbb{E}_Q \left[\int_t^T u(s) \rho(\hat{y}_s^+) ds \middle| \mathcal{F}_t \right] \quad dP\text{-a.s.}$$

Thus, applying Lemma 2.1 with $u_n(t) \equiv \hat{y}_t^+$, $b_n \equiv 0$, $P_n \equiv Q$, $\beta(s) = u(s)$ and $\psi(u) = \rho(u)$ yields that for each $t \in [0, T]$,

$$\hat{y}_t^+ = \lim_{n \rightarrow \infty} \hat{y}_t^+ = \lim_{n \rightarrow \infty} u_n(t) = 0 \quad dP\text{-a.s.}$$

That is to say, for each $t \in [0, T]$, $y_t \leq y'_t$ dP -a.s. The proof is complete. $\square$

Theorem 4.2. Assume that $0 < T \leq +\infty$, $\xi, \xi' \in L^\infty(\Omega, \mathcal{F}_T, P)$, both g and g' satisfy (3A2), and (y, z) and (y', z') are, respectively, a bounded solution of BSDE(ξ, g) and BSDE(ξ', g'). If $\xi \leq \xi'$ dP -a.s. and one of the following two statements is satisfied:

- (i) g satisfies (2A1) and (4A2), and (8) holds true;
- (ii) g' satisfies (2A1) and (4A2), and (9) holds true,

then for each $t \in [0, T]$,

$$y_t \leq y'_t \quad dP\text{-a.s.}$$

Proof. The proof will be split into four steps.

First step: Suppose that both g and g' satisfy (3A2), $\xi \leq \xi'$ dP -a.s., g also satisfies (2A1) and is convex with respect to z , and (8) holds true. We further assume that both y and y' are non-positive.

For each $n \geq 2$, let us set

$$\hat{y}_t^n = y_t - \frac{n-1}{n} y'_t, \quad \hat{z}_t^n = z_t - \frac{n-1}{n} z'_t.$$

Note that

$$\left(\xi - \frac{n-1}{n} \xi' \right)^+ \leq (\xi - \xi')^+ = 0$$

due to the fact that $\xi' \leq 0$. Tanaka's formula yields that for $t \in [0, T]$,

$$(\hat{y}_t^n)^+ \leq \int_t^T \mathbb{1}_{\hat{y}_s^n > 0} \left[g(s, y_s, z_s) - \frac{n-1}{n} g'(s, y'_s, z'_s) \right] ds - \int_t^T \mathbb{1}_{\hat{y}_s^n > 0} \hat{z}_s^n \cdot dB_s. \quad (38)$$

First of all, in view of $y'_s \leq 0$ and then

$$\frac{n-1}{n} y'_s \geq y'_s,$$

it follows from (8) that $\mathbb{1}_{\hat{y}_s^n > 0} (g(s, y'_s, z'_s) - g'(s, y'_s, z'_s))$ is non-positive. Then we have

$$\begin{aligned} & \mathbb{1}_{\hat{y}_s^n > 0} \left(g(s, y_s, z_s) - \frac{n-1}{n} g'(s, y'_s, z'_s) \right) \\ &= \mathbb{1}_{\hat{y}_s^n > 0} \left(g(s, y_s, z_s) - \frac{n-1}{n} g(s, y'_s, z'_s) \right) \\ & \quad + \mathbb{1}_{\hat{y}_s^n > 0} \frac{n-1}{n} (g(s, y'_s, z'_s) - g'(s, y'_s, z'_s)) \\ & \leq \mathbb{1}_{\hat{y}_s^n > 0} (g(s, y_s, z_s) - g(s, y'_s, z_s)) + \mathbb{1}_{\hat{y}_s^n > 0} \left(g(s, y'_s, z_s) - \frac{n-1}{n} g(s, y'_s, z'_s) \right). \end{aligned} \quad (39)$$

By (2A1) we can obtain that, in view of $y'_s \leq 0$,

$$\begin{aligned} & \mathbb{1}_{\hat{y}_s^n > 0} (g(s, y_s, z_s) - g(s, y'_s, z_s)) \\ &= \mathbb{1}_{\hat{y}_s^n > 0} \left[\left(g(s, y_s, z_s) - g\left(s, \frac{n-1}{n} y'_s, z_s\right) \right) \right. \\ & \quad \left. + \left(g\left(s, \frac{n-1}{n} y'_s, z_s\right) - g(s, y'_s, z_s) \right) \right] \\ & \leq u(s) \rho((\hat{y}_s^n)^+) + u(s) \rho\left(\frac{-y'_s}{n}\right). \end{aligned} \quad (40)$$

Furthermore, since g is convex with respect to z , and satisfies (3A2), we have

$$\begin{aligned} g(s, y'_s, z_s) &= g\left(s, y'_s, \frac{n-1}{n} z'_s + \frac{1}{n} (nz_s - (n-1)z'_s)\right) \\ &\leq \frac{n-1}{n} g(s, y'_s, z'_s) + \frac{1}{n} g(s, y'_s, n\hat{z}_s^n) \\ &\leq \frac{n-1}{n} g(s, y'_s, z'_s) + \frac{\bar{u}(s)\bar{\varphi}(y'_s)}{n} + n\bar{h}(y'_s)|\hat{z}_s^n|^2. \end{aligned} \quad (41)$$

Thus, in view of the fact that $-k \leq y'_s \leq 0$ for some positive constant $k > 0$, combining (38)–(41) yields that

$$\begin{aligned} (\hat{y}_t^n)^+ &\leq a_n + \int_t^T (u(s) \rho((\hat{y}_s^n)^+) + n\gamma \mathbb{1}_{\hat{y}_s^n > 0} |\hat{z}_s^n|^2) ds - \int_t^T \mathbb{1}_{\hat{y}_s^n > 0} \hat{z}_s^n \cdot dB_s \\ &= a_n + \int_t^T u(s) \rho((\hat{y}_s^n)^+) ds - \int_t^T \mathbb{1}_{\hat{y}_s^n > 0} \hat{z}_s^n \cdot [dB_s - n\gamma \hat{z}_s^n ds], \quad t \in [0, T], \end{aligned} \quad (42)$$

where $\gamma := \max_{x \in [-k, 0]} \bar{h}(x)$ and

$$a_n := \rho\left(\frac{k}{n}\right) \int_0^T u(s) \, ds + \frac{\max_{x \in [-k, 0]} \bar{\varphi}(x)}{n} \int_0^T \bar{u}(s) \, ds \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

In the sequel, since both g and g' satisfy (3A2), and both (y, z) and (y', z') are bounded solutions, it follows from Lemma 4.1 that for each $n \geq 2$, the process

$$N_t^n := \int_0^t n \gamma \hat{z}_s^n \cdot dB_s, \quad t \in [0, T]$$

is a BMO-martingale under P . Then the stochastic exponential

$$\mathcal{E}(N^n)_t = \exp\left(N_t^n - \frac{1}{2} \langle N^n \rangle_t\right), \quad t \in [0, T]$$

of N^n is a uniformly integrable martingale. Now, let us define by P_n the probability measure on $(\Omega, \mathcal{F}_T)$ given by

$$\frac{dP_n}{dP} := \mathcal{E}(N^n)_T.$$

Then, noticing that

$$\bar{N}_t^n := \int_0^t \mathbb{1}_{\hat{y}_s^n > 0} \hat{z}_s^n \cdot dB_s, \quad t \in [0, T]$$

is also a BMO-martingale under P , we know that the process

$$\int_0^t \mathbb{1}_{\hat{y}_s^n > 0} \hat{z}_s^n \cdot [dB_s - n \gamma \hat{z}_s^n \, ds], \quad t \in [0, T],$$

the Girsanov's transform of $\bar{N}^n$, is a BMO-martingale under P_n . Let $\mathbb{E}_n[X|\mathcal{F}_t]$ represent the conditional expectation of the random variable X with respect to $\mathcal{F}_t$ under P_n . Taking the conditional expectation with respect to $\mathcal{F}_t$ under P_n in (42) yields that for each $t \in [0, T]$,

$$(\hat{y}_t^n)^+ \leq a_n + \mathbb{E}_n \left[\int_t^T u(s) \rho((\hat{y}_s^n)^+) \, ds \middle| \mathcal{F}_t \right] \quad dP\text{-a.s.}$$

Thus, applying Lemma 2.1 with $u_n(t) = (\hat{y}_t^n)^+$, $b_n = a_n$, $\beta(s) = u(s)$ and $\psi(u) = \rho(u)$ yields that for each $t \in [0, T]$,

$$(y_t - y'_t)^+ = \lim_{n \rightarrow \infty} (\hat{y}_t^n)^+ = \lim_{n \rightarrow \infty} u_n(t) = 0 \quad dP\text{-a.s.}$$

That is to say, for each $t \in [0, T]$, $y_t \leq y'_t$ dP -a.s.

Second step: In this step, we will eliminate the condition that both y and y' are non-positive required in the first step.

Indeed, note that both y and y' are bounded processes. We can assume that there exists a constant $k > 0$ such that for each $t \in [0, T]$, $|y_t| + |y'_t| \leq k$ dP -a.s. Let $\bar{y}_t := y_t - k$ and $\bar{y}'_t := y'_t - k$, then $(\bar{y}_t, z_t)_{t \in [0, T]}$ and $(\bar{y}'_t, z'_t)_{t \in [0, T]}$ are, respectively, a bounded solution of BSDE $(\bar{\xi}, \bar{g})$ and BSDE $(\bar{\xi}', \bar{g}')$, where $\bar{\xi} := \xi - k$, $\bar{\xi}' := \xi' - k$ and for each $(\omega, t, y, z) \in \Omega \times [0, T] \times \mathbb{R} \times \mathbb{R}^d$,

$$\bar{g}(\omega, t, y, z) := g(\omega, t, y + k, z), \quad \bar{g}'(\omega, t, y, z) := g'(\omega, t, y + k, z).$$

It is not difficult to verify that both $\bar{g}$ and $\bar{g}'$ satisfy (3A2) but $\bar{\varphi}(u)$ and $\bar{h}(u)$ are replaced by $\bar{\varphi}(u+k)$ and $\bar{h}(u+k)$ respectively, $\bar{\xi} \leq \bar{\xi}'$ dP -a.s., $\bar{g}$ satisfies (2A1), $dP \times dt$ -a.e.,

$$\begin{aligned}\mathbb{1}_{\bar{y}_t > \bar{y}'_t} \bar{g}(t, \bar{y}'_t, z'_t) &= \mathbb{1}_{y_t > y'_t} g(t, y'_t, z'_t) \\ &\leq \mathbb{1}_{y_t > y'_t} g'(t, y'_t, z'_t) \\ &= \mathbb{1}_{\bar{y}_t > \bar{y}'_t} \bar{g}'(t, \bar{y}'_t, z'_t),\end{aligned}$$

$\bar{g}$ is convex with respect to z , and both $\bar{y}$ and $\bar{y}'$ are non-positive. Thus, by the first step we can conclude that for each $t \in [0, T]$,

$$y_t - k = \bar{y}_t \leq \bar{y}'_t = y'_t - k \quad dP\text{-a.s.},$$

which is the desired result.

Third step: Suppose that both g and g' satisfy (3A2), $\xi \leq \xi'$ dP -a.s., g' also satisfies (2A1) and is convex with respect to z , and (9) holds true. Then in view of (9) we can replace (39) by the following inequality

$$\begin{aligned}&\mathbb{1}_{\hat{y}_s^n > 0} \left(g(s, y_s, z_s) - \frac{n-1}{n} g'(s, y'_s, z'_s) \right) \\ &= \mathbb{1}_{\hat{y}_s^n > 0} (g(s, y_s, z_s) - g'(s, y_s, z_s)) + \mathbb{1}_{\hat{y}_s^n > 0} \left(g'(s, y_s, z_s) - \frac{n-1}{n} g'(s, y'_s, z'_s) \right) \\ &\leq \mathbb{1}_{\hat{y}_s^n > 0} (g'(s, y_s, z_s) - g'(s, y'_s, z'_s)) + \mathbb{1}_{\hat{y}_s^n > 0} \left(g'(s, y'_s, z_s) - \frac{n-1}{n} g'(s, y'_s, z'_s) \right),\end{aligned}$$

and then make use of the assumptions on g' to conclude that (42) holds still true provided that $y'_t \leq 0$. Thus, the similar argument to the previous two steps will give the desired conclusion.

Fourth step: Suppose that both g and g' satisfy (3A2), $\xi \leq \xi'$ dP -a.s., g also satisfies (2A1) and is concave with respect to z , and (8) holds true.

Let us set

$$\tilde{\xi}' := -\xi', \quad \tilde{y}'_t := -y'_t, \quad \tilde{z}'_t := -z'_t, \quad \tilde{g}'(t, y, z) := -g'(t, -y, -z)$$

and

$$\tilde{\xi} := -\xi, \quad \tilde{y}_t := -y_t, \quad \tilde{z}_t := -z_t, \quad \tilde{g}(t, y, z) := -g(t, -y, -z).$$

Then $(\tilde{y}'_t, \tilde{z}'_t)_{t \in [0, T]}$ and $(\tilde{y}_t, \tilde{z}_t)_{t \in [0, T]}$ are, respectively, a bounded solution of BSDE $(\tilde{\xi}', \tilde{g}')$ and BSDE $(\tilde{\xi}, \tilde{g})$. And, it is easy to verify that both $\tilde{g}'$ and $\tilde{g}$ satisfy (3A2), $\tilde{\xi}' \leq \tilde{\xi}$ dP -a.s., $\tilde{g}$ satisfies (2A1) and is convex with respect to z , and $dP \times dt$ -a.e.,

$$\begin{aligned}\mathbb{1}_{\tilde{y}'_t > \tilde{y}_t} \tilde{g}'(t, \tilde{y}'_t, \tilde{z}'_t) &= -\mathbb{1}_{y_t > y'_t} g'(t, y'_t, z'_t) \\ &\leq -\mathbb{1}_{y_t > y'_t} g(t, y'_t, z'_t) \\ &= \mathbb{1}_{\tilde{y}'_t > \tilde{y}_t} \tilde{g}(t, \tilde{y}'_t, \tilde{z}'_t).\end{aligned}$$

Thus, by the third step we know that for each $t \in [0, T]$,

$$-y'_t = \tilde{y}'_t \leq \tilde{y}_t = -y_t \quad dP\text{-a.s.},$$

which is the desired result. In the same way, we can prove the remainder case that both g and g' satisfy (3A2), $\xi \leq \xi'$ dP -a.s., g' also satisfies (2A1) and is concave with respect to z , and (9) holds true. **Theorem 4.2** is then proved. $\square$

By virtue of [Theorems 2.1, 3.1, 4.1](#) and [4.2](#), we can obtain the following existence and uniqueness result for bounded solutions of BSDEs.

Theorem 4.3. Assume that $0 < T \leq +\infty$ and that g satisfies (2A3), (3A2) and (2A1). Furthermore, we also assume that one of three assumptions (2A2), (4A1) and (4A2) holds true for g . Then for each $\xi \in L^\infty(\Omega, \mathcal{F}_T, P)$, BSDE(ξ, g) has a unique bounded solution (Y, Z) . Moreover, for each $t \in [0, T]$, we have

$$L_0 \leq L_t \leq Y_t \leq U_t \leq U_0 \quad dP\text{-a.s.},$$

where (L, U) are the unique solutions of [\(27\)](#) and [\(28\)](#) but $u(t)$ is replaced by $u(t) + \bar{u}(t)$, and

$$l(x) := \rho(|x|) + \bar{\varphi}(0) + 1, \quad \forall x \in \mathbb{R}. \quad (43)$$

Proof. It follows from (2A1) and (3A2) that $dP \times dt$ -a.e., for each $(y, z) \in \mathbb{R} \times \mathbb{R}^d$,

$$g(\omega, t, y, z) \operatorname{sgn}(y) \leq u(t)\rho(|y|) + |g(t, 0, z)| \leq [u(t) + \bar{u}(t)]l(y) + \bar{h}(0)|z|^2,$$

where $l(x)$ is defined in [\(43\)](#). Note that ρ is of linear growth. We can deduce that $l(\cdot)$ belongs to $\mathcal{L}(\mathbb{R}; \mathbb{R}^+)$ and then (3A1) holds true for the generator g . Thus, the existence of a bounded solution of BSDE(ξ, g) follows from [Theorem 3.1](#). Finally, the uniqueness follows directly from [Theorems 2.1, 4.1](#) and [4.2](#). The proof is then completed. $\square$

Remark 4.1. By the proof of [Theorem 2.1](#) and [Theorems 4.1–4.3](#) we can conclude that if assumptions (2A2), (4A1) and (4A2) hold true only when $y \in [L_0, U_0]$, then the conclusions of [Theorem 4.3](#) hold still true.

5. Existence and uniqueness of L^p ($p > 1$) solutions

In this section, we will fix a real number $p > 1$ and establish two existence results, two comparison theorems and an existence and uniqueness result for the (maximal and minimal) L^p solutions of BSDEs. As before, let us first introduce the following assumptions on the generator g , where we also assume that $0 < T \leq +\infty$.

(5A1) There exist two functions $u(\cdot) \in L^1([0, T]; \mathbb{R}_+)$, $v(\cdot) \in L^2([0, T]; \mathbb{R}_+)$ and a process $f_t \in L^p(\Omega; L^1([0, T]; \mathbb{R}_+))$ such that $dP \times dt$ -a.e.,

$$g(\omega, t, y, z) \operatorname{sgn}(y) \leq f_t(\omega) + u(t)|y| + v(t)|z|, \quad \forall y, z.$$

(5A2) There exist two functions $\bar{u}(\cdot) \in L^1([0, T]; \mathbb{R}_+)$, $\bar{v}(\cdot) \in L^2([0, T]; \mathbb{R}_+)$ and a process $\bar{f}_t \in L^p(\Omega; L^1([0, T]; \mathbb{R}_+))$ such that $dP \times dt$ -a.e.,

$$\forall y \leq 0, \forall z \in \mathbb{R}^d, \quad g(\omega, t, y, z) \leq \bar{f}_t(\omega) + \bar{u}(t)|y| + \bar{v}(t)|z|.$$

(5A3) There exist two functions $\bar{u}(\cdot) \in L^1([0, T]; \mathbb{R}_+)$, $\bar{v}(\cdot) \in L^2([0, T]; \mathbb{R}_+)$ and a process $\bar{f}_t \in L^p(\Omega; L^1([0, T]; \mathbb{R}_+))$ such that $dP \times dt$ -a.e.,

$$\forall y \geq 0, \forall z \in \mathbb{R}^d, \quad -g(\omega, t, y, z) \leq \bar{f}_t(\omega) + \bar{u}(t)|y| + \bar{v}(t)|z|.$$

(5A4) There exist two functions $u(\cdot) \in L^1([0, T]; \mathbb{R}_+)$, $v(\cdot) \in L^2([0, T]; \mathbb{R}_+)$ and a process $f_t \in L^p(\Omega; L^1([0, T]; \mathbb{R}_+))$ such that $dP \times dt$ -a.e.,

$$|g(\omega, t, y, z)| \leq f_t(\omega) + u(t)|y| + v(t)|z|, \quad \forall y, z.$$

(5A5) There exist a nonnegative $(\mathcal{F}_t)$ -progressively measurable process $(\check{u}_t)_{t \in [0, T]}$ with dP -a.s., $\int_0^T \check{u}_t(\omega) dt < +\infty$, and two continuous functions $\check{\varphi}(\cdot), \check{h}(\cdot) : \mathbb{R} \mapsto \mathbb{R}_+$ such that $dP \times dt$ -a.e.,

$$|g(\omega, t, y, z)| \leq \check{u}_t(\omega) \check{\varphi}(y) + \check{h}(y) |z|^2, \quad \forall y, z.$$

Remark 5.1. It is clear that $(5A4) \iff (5A1) + (5A2) + (5A3)$, $(5A4) \implies (5A5)$, and $(3A2) \implies (5A5)$.

The following lemma will be used in this section, which comes from Proposition 2.3 in Fan and Jiang [17].

Lemma 5.1. Assume that $0 < T \leq +\infty$, g is a generator of BSDEs and the process $g(t, 0, 0) \in L^p(\Omega; L^1([0, T]; \mathbb{R}_+))$. Assume further that there exist two functions $u(\cdot) \in L^1([0, T]; \mathbb{R}_+)$ and $v(\cdot) \in L^2([0, T]; \mathbb{R}_+)$ such that $dP \times dt$ -a.e.,

$$|g(\omega, t, y_1, z_1) - g(\omega, t, y_2, z_2)| \leq u(t)|y_1 - y_2| + v(t)|z_1 - z_2|, \quad \forall y_1, y_2, z_1, z_2. \quad (44)$$

Then for each $\xi \in L^p(\Omega, \mathcal{F}_T, P)$, BSDE(ξ, g) has a unique L^p solution.

By Theorem 2.1 and Remark 2.2, we can establish the following lemma.

Lemma 5.2. Assume that $0 < T \leq +\infty$, $\xi \in L^p(\Omega, \mathcal{F}_T, P)$, and g satisfies (2A3) and (5A1). Let $(y_t, z_t)_{t \in [0, T]}$ be any L^p solution of BSDE(ξ, g) and $(y'_t, z'_t)_{t \in [0, T]}$ the unique L^p solution of BSDE($|\xi|, g'$) (recall Lemma 5.1), where

$$g'(\omega, t, y, z) := f_t(\omega) + u(t)|y| + v(t)|z|, \quad \forall \omega, t, y, z.$$

Then for each $t \in [0, T]$, we have

$$|y_t| \leq y'_t \quad dP\text{-a.s.}$$

Proof. Note that $|\xi| \geq 0$ dP -a.s. and $g'(t, 0, 0) = f_t \geq 0$ $dP \times dt$ -a.e. By Theorem 2.1 we know that for each $t \in [0, T]$,

$$y'_t \geq 0 \quad dP\text{-a.s.} \quad (45)$$

It follows from (5A1) that $dP \times dt$ -a.e.,

$$\forall y > 0, \forall z \in \mathbb{R}^d, \quad g(\omega, t, y, z) \leq g'(\omega, t, y, z). \quad (46)$$

Thus, in view of (45), (46), Remark 2.2 and the fact that $\xi \leq |\xi|$ dP -a.s., by Theorem 2.1 we deduce that for each $t \in [0, T]$,

$$y_t \leq y'_t \quad dP\text{-a.s.}$$

Furthermore, by (5A1) we can also deduce that $dP \times dt$ -a.e.,

$$\forall y < 0, \forall z \in \mathbb{R}^d, \quad -g'(\omega, t, y, z) \leq g(\omega, t, y, z). \quad (47)$$

On the other hand, it is not difficult to verify that $(-y'_t, -z'_t)_{t \in [0, T]}$ is the unique L^p solution of BSDE($-|\xi|, -g'$). Then, in view of (45), (47), Remark 2.2 and the fact that $-|\xi| \leq \xi$ dP -a.s., by Theorem 2.1 we can deduce that for each $t \in [0, T]$,

$$-y'_t \leq y_t \quad dP\text{-a.s.}$$

Thus, we have completed the proof of Lemma 5.2. $\square$

The following [Theorem 5.1](#) establishes an existence result for L^p solutions of BSDEs, which is one of main results in this section. It improves considerably the corresponding existence results obtained respectively in Lepeltier and San Martín [27], Chen [10] and Briand, Lepeltier and San Martín [8] even for the case of finite time horizon.

Theorem 5.1. Assume that $0 < T \leq +\infty$ and g satisfies (2A3), (3A2) and (5A1) with $f_t \in L^1([0, T]; \mathbb{R}_+)$. Then for each $\xi \in L^p(\Omega, \mathcal{F}_T, P)$, BSDE(ξ, g) has an L^p solution.

Proof. Let us first assume that ξ is nonnegative. Note first that (5A1) with $f_t \in L^1([0, T]; \mathbb{R}_+)$ can imply (3A1). Indeed, if g satisfies (5A1), then $dP \times dt$ -a.e.,

$$g(\omega, t, y, z) \operatorname{sgn}(y) \leq \tilde{u}(t)\tilde{l}(y) + |z|^2, \quad \forall y, z,$$

where $\tilde{u}(t) := f_t + u(t) + v^2(t) \in L^1([0, T]; \mathbb{R}_+)$ and $\tilde{l}(x) := 1 + |x| \in \mathcal{L}(\mathbb{R}; \mathbb{R}^+)$.

It follows from [Theorem 3.1](#) that BSDE(ξ_n, g) has a maximal bounded solution $(y_t^n, z_t^n)_{t \in [0, T]}$ for each $n \geq 1$, where $\xi_n := \xi \wedge n$. Furthermore, in view of $\xi_n \leq \xi_{n+1}$, by [Corollary 3.1](#) we also know that for each $t \in [0, T]$, $(y_t^n)_{n=1}^{+\infty}$ is nondecreasing.

In the sequel, for each $(\omega, t, y, z) \in \Omega \times [0, T] \times \mathbb{R} \times \mathbb{R}^d$, we define

$$g'(\omega, t, y, z) := f_t + u(t)|y| + v(t)|z|.$$

It follows from [Lemma 5.1](#) that for each $n \geq 1$, BSDE($|\xi_n|, g'$) has a unique L^p solution $(^n y_t', ^n z_t')_{t \in [0, T]}$. Furthermore, in view of (5A1), by [Lemma 5.2](#) we can conclude that for each $t \in [0, T]$ and each $n \geq 1$, $|y_t^n| \leq ^n y_t'$ dP -a.s.

On the other hand, in view of $\xi \in L^p(\Omega, \mathcal{F}_T, P)$, it follows from [Lemma 5.1](#) that BSDE($|\xi|, g'$) has also a unique L^p solution $(y_t', z_t')_{t \in [0, T]}$. Then, in view of $|\xi_n| \leq |\xi|$, by [Theorem 2.1](#) we know that for each $t \in [0, T]$ and each $n \geq 1$, $^n y_t' \leq y_t'$ dP -a.s.

As a result, we have proved that for each $t \in [0, T]$ and $n \geq 1$,

$$-y_t' \leq y_t^n \leq y_t^{n+1} \leq y_t' \quad dP\text{-a.s.} \quad (48)$$

We define $y = \lim_{n \rightarrow \infty} y^n$, then

$$\forall t \in [0, T], \quad |y_t| \leq |y_t'| = y_t' \quad dP\text{-a.s.}$$

In the sequel, we will use the localization procedure used in Briand and Hu [6] to construct the desired solution. For each $k \geq 1$, let us introduce the following stopping time:

$$\tau_k = \inf\{t \in [0, T] : |y_t'| \geq k\} \wedge T.$$

Then $(y_k^n(t), z_k^n(t)) := (y_{t \wedge \tau_k}^n, z_{t \wedge \tau_k}^n \mathbb{1}_{t \leq \tau_k})$ solves the following BSDE:

$$y_k^n(t) = y_{\tau_k}^n + \int_t^T \mathbb{1}_{s \leq \tau_k} g(s, y_k^n(s), z_k^n(s)) ds - \int_t^T z_k^n(s) \cdot dB_s. \quad (49)$$

It is very important to observe that y_k^n is nondecreasing in n and that, from the definition of τ_k and inequality (48),

$$\sup_{n \geq 1} \sup_{t \in [0, T]} \|y_k^n(t)\|_\infty \leq k.$$

Furthermore, by (3A2) we know that $dP \times dt$ -a.e., for each $(y, z) \in [-k, k] \times \mathbb{R}^d$,

$$|g(\omega, t, y, z)| \leq \bar{u}(t) \left(\max_{x \in [-k, k]} \bar{\varphi}(x) \right) + \left(\max_{x \in [-k, k]} \bar{h}(x) \right) |z|^2.$$

Thus, arguing as in the last second paragraph of the proof of Lemma 3.3, we can take the limit with respect to n (k being fixed) in (49) in the space $\mathcal{S}^2 \times \mathcal{M}^2$. In particular, setting $y_k(t) = \sup_{n \geq 1} y_k^n(t)$, we know that $y_k(\cdot)$ is continuous and that there exists a process $z_k(t) \in \mathcal{M}^2$ such that $\lim_{n \rightarrow \infty} z_k^n(t) = z_k(t)$ in $\mathcal{M}^2$ and $(y_k(t), z_k(t))$ solves the following BSDE

$$y_k(t) = \sup_{n \geq 1} y_{\tau_k}^n + \int_t^T \mathbb{1}_{s \leq \tau_k} g(s, y_k(s), z_k(s)) ds - \int_t^T z_k(s) \cdot dB_s. \quad (50)$$

Since $\tau_k \leq \tau_{k+1}$, it follows from the definitions of $y_k(\cdot)$, $z_k(\cdot)$ and y , that

$$y_{t \wedge \tau_k} = y_{k+1}(t \wedge \tau_k) = y_k(t) = \sup_{n \geq 1} y_{t \wedge \tau_k}^n, \quad z_{k+1}(t) \mathbb{1}_{t \leq \tau_k} = z_k(t) = \lim_{n \rightarrow \infty} z_t^n \mathbb{1}_{t \leq \tau_k}.$$

Thus, since $y_k(\cdot)$ are continuous processes and moreover dP -a.s., $\tau_k = T$ for k large enough, we know that y is continuous on $[0, T]$. Then we define z on $(0, T)$ by setting

$$z_t = z_k(t), \quad \text{if } t \in (0, \tau_k),$$

so that $z_t \mathbb{1}_{t \leq \tau_k} = z_k(t) \mathbb{1}_{t \leq \tau_k} = z_k(t)$ and (49) can be rewritten as

$$y_{t \wedge \tau_k} = y_{\tau_k} + \int_{t \wedge \tau_k}^{\tau_k} g(s, y_s, z_s) ds - \int_{t \wedge \tau_k}^{\tau_k} z_s \cdot dB_s. \quad (51)$$

Furthermore, we have

$$\begin{aligned} P\left(\int_0^T |z_s|^2 ds = \infty\right) &= P\left(\int_0^T |z_s|^2 ds = \infty, \tau_k = T\right) + P\left(\int_0^T |z_s|^2 ds = \infty, \tau_k < T\right) \\ &\leq P\left(\int_0^{\tau_k} |z_k(s)|^2 ds = \infty\right) + P(\tau_k < T), \end{aligned}$$

and we deduce, since $\tau_k \uparrow T$, that

$$\int_0^T |z_s|^2 ds < \infty \quad dP\text{-a.s.}$$

Thus, letting $k \rightarrow \infty$ in (51), we can deduce that (y, z) is a solution of BSDE(ξ, g). Furthermore, by (48) we know that $y \in \mathcal{S}^p$, and then arguing as in Lemma 3.1 of Briand, Delyon, Hu, Pardoux and Stoica [5], in view of (5A1), we can deduce that $z \in \mathcal{M}^p$. That is to say, (y, z) is also an L^p solution of BSDE(ξ, g).

In the general case, we can use a double approximation: $\xi_{n,p} := \xi^+ \wedge n - \xi^- \wedge p$ as in Briand and Hu [6] and Briand and Hu [7]. The proof is then complete. $\square$

The following Theorem 5.2 gives a new existence result on the maximal and minimal L^p solution of BSDEs. In view of Remark 5.1, we know that it generalizes Theorem 1 in Fan, Jiang and Tian [19] (see Lemma 2.2) even for the case of L^2 solutions.

Theorem 5.2. Assume that $0 < T \leq +\infty$ and g satisfies (2A3), (5A1) and (5A5). Assume further that g satisfies (5A2) (resp. (5A3)). Then for each $\xi \in L^p(\Omega, \mathcal{F}_T, P)$, BSDE(ξ, g) has a maximal (resp. minimal) L^p solution.

Proof. We only prove the case of maximal solution, the other case is similar. Assume now that $\xi \in L^p(\Omega, \mathcal{F}_T, P)$ and that g satisfies (2A3), (5A1), (5A5) and (5A2). By (5A1) and (5A2) we know that $dP \times dt$ -a.e.,

$$g(\omega, t, y, z) \leq (f_t(\omega) + \bar{f}_t(\omega)) + (u(t) + \bar{u}(t))|y| + (v(t) + \bar{v}(t))|z|, \quad \forall y, z.$$

Then for each $n \geq 1$ and $(\omega, t, y, z) \in \Omega \times [0, T] \times \mathbb{R} \times \mathbb{R}^d$, we can define

$$g_n(\omega, t, y, z) := \sup_{(u,v) \in \mathbb{R}^{1+d}} \{g(\omega, t, u, v) - n(u(t) + \bar{u}(t))|y - u| - n(v(t) + \bar{v}(t))|z - v|\}. \quad (52)$$

Arguing as in Fan, Jiang and Tian [19], we can also conclude that the sequence of functions g_n is well defined for each $n \geq 1$, and it satisfies, $dP \times dt$ -a.e.,

- (i) $\forall y, z, g_n(\omega, t, y, z) \leq (f_t(\omega) + \bar{f}_t(\omega)) + (u(t) + \bar{u}(t))|y| + (v(t) + \bar{v}(t))|z|$.
- (ii) $\forall y, z, g_n(\omega, t, y, z)$ non-increases in n .
- (iii) $\forall y_1, y_2, z_1, z_2$, we have

$$|g_n(\omega, t, y_1, z_1) - g_n(\omega, t, y_2, z_2)| \leq n(u(t) + \bar{u}(t))|y_1 - y_2| + n(v(t) + \bar{v}(t))|z_1 - z_2|.$$

- (iv) If $(y_n, z_n) \rightarrow (y, z)$, then $g_n(\omega, t, y_n, z_n) \rightarrow g(\omega, t, y, z)$.

Furthermore, it follows from (52) and (5A1) that for each $n \geq 1$, $dP \times dt$ -a.e.,

$$\forall y < 0, \forall z \in \mathbb{R}^d, \quad g'(\omega, t, y, z) \leq g(\omega, t, y, z) \leq g_n(\omega, t, y, z), \quad (53)$$

where

$$g'(\omega, t, y, z) := -f_t(\omega) - u(t)|y| - v(t)|z|, \quad \forall \omega, t, y, z.$$

Note that (iii) and $|g_n(t, 0, 0)| \in L^p(\Omega; L^1([0, T]; \mathbb{R}_+))$ by (i) and (53). It follows from Lemma 5.1 that BSDE (ξ, g_n) has a unique L^p solution $(y_t^n, z_t^n)_{t \in [0, T]}$ for each $n \geq 1$. In view of (ii) and (iii), by Theorem 2.1 we know that for each $t \in [0, T]$, the sequel $(y_t^n)_{n=1}^{+\infty}$ is non-increasing. On the other hand, let (y_t', z_t') be the unique L^p solution of BSDE $(-|\xi|, g')$ by Lemma 5.1. Note that $y_t' \leq 0$ by Theorem 2.1. In view of (53), Remark 2.2 and the fact that $-|\xi| \leq \xi$ dP -a.s., by Theorem 2.1 we know that for each $t \in [0, T]$ and each $n \geq 1$,

$$y_t' \leq y_t^{n+1} \leq y_t^n \leq y_t^1 \quad dP\text{-a.s.} \quad (54)$$

In the sequel, for each $k \geq 1$, let us introduce the following stopping time:

$$\tau_k = \inf \left\{ t \in [0, T] : |y_t'| + |y_t^1| + \int_0^t (\check{u}_s + f_s + \bar{f}_s) ds \geq k \right\} \wedge T.$$

Then $(y_k^n(t), z_k^n(t)) := (y_{t \wedge \tau_k}^n, z_{t \wedge \tau_k}^n)$ solves the following BSDE:

$$y_k^n(t) = y_{\tau_k}^n + \int_t^T \mathbb{1}_{s \leq \tau_k} g_n(s, y_k^n(s), z_k^n(s)) ds - \int_t^T z_k^n(s) \cdot dB_s.$$

In view of (i) and the facts that $g_n \geq g$ and g satisfies (5A5), we know that $dP \times dt$ -a.e., for each $(y, z) \in [-k, k] \times \mathbb{R}^d$,

$$\begin{aligned} \sup_{n \geq 1} |g_n(\omega, t, y, z)| &\leq \check{u}_t(\omega) \left(\max_{x \in [-k, k]} \check{\varphi}(x) \right) + f_t(\omega) + \bar{f}_t(\omega) + k(u(t) + \bar{u}(t)) \\ &\quad + (v(t) + \bar{v}(t))^2 + \left(\max_{x \in [-k, k]} \check{h}(x) + 1 \right) |z|^2. \end{aligned}$$

Thus, in view of (54) and the definition of τ_k , by a similar argument to Lemma 4.1 we can deduce that $(y_k^n(t), z_k^n(t)) \in \mathcal{S}^\infty \times \mathcal{M}^2$ for each $n \geq 1$ and $k \geq 1$. Furthermore, in view of (iv), (54) and

the above inequality together with the definition of τ_k , arguing as in the proof of [Theorem 5.1](#), we can define $y = \lim_{n \rightarrow \infty} y^n$ and use the localization procedure to obtain a process z such that (y, z) is an L^p solution of $\text{BSDE}(\xi, g)$.

Finally, it remains to show that (y, z) is the maximal one among all L^p solutions of $\text{BSDE}(\xi, g)$. Indeed, let $(\hat{y}, \hat{z})$ be any L^p solutions of $\text{BSDE}(\xi, g)$. In view of (iii) and the fact that $g_n \geq g$, by [Theorem 2.1](#) we can obtain that for each $n \geq 1$ and each $t \in [0, T]$,

$$\hat{y}_t \leq y_t^n \quad dP\text{-a.s.}$$

Letting $n \rightarrow \infty$ yields the desired result. The proof is then completed. $\square$

By [Theorem 5.2](#) and [Remark 5.1](#), the following corollary is immediate. It generalizes [Lemma 2.2](#) to the case of L^p solutions.

Corollary 5.1. Assume that $0 < T \leq +\infty$ and g satisfies (2A3) and (5A4). Then for each $\xi \in L^p(\Omega, \mathcal{F}_T, P)$, $\text{BSDE}(\xi, g)$ has both a maximal and a minimal L^p solution.

Similar to [Theorems 2.2](#) and [2.3](#), by virtue of [Theorem 2.1](#) we can prove the following [Theorems 5.3](#) and [5.4](#), which establish the comparison theorems on the maximal and minimal L^p solutions of BSDEs under the assumptions of [Theorem 5.2](#). In view of [Remark 5.1](#), they generalize [Theorems 2.2](#) and [2.3](#) even for the case of L^2 solutions.

Theorem 5.3. Assume that $0 < T \leq +\infty$, $\xi, \xi' \in L^p(\Omega, \mathcal{F}_T, P)$, g and g' are two generators of BSDEs, and (y, z) is any L^p solution of $\text{BSDE}(\xi, g)$. Assume further that g' satisfies (2A3), (5A1), (5A2) and (5A5), and (y', z') is the maximal L^p solution of $\text{BSDE}(\xi', g')$ (recall [Theorem 5.2](#)). If $\xi \leq \xi'$ dP -a.s. and (9) holds true, then for each $t \in [0, T]$,

$$y_t \leq y'_t \quad dP\text{-a.s.}$$

Theorem 5.4. Assume that $0 < T \leq +\infty$, $\xi, \xi' \in L^p(\Omega, \mathcal{F}_T, P)$, g and g' are two generators of BSDEs, and (y', z') is any L^p solution of $\text{BSDE}(\xi', g')$. Assume further that g satisfies (2A3), (5A1), (5A3) and (5A5), and (y, z) is the minimal L^p solution of $\text{BSDE}(\xi, g)$ (recall [Theorem 5.2](#)). If $\xi \leq \xi'$ dP -a.s. and (8) holds true, then for each $t \in [0, T]$,

$$y_t \leq y'_t \quad dP\text{-a.s.}$$

Corollary 5.2. Assume that $0 < T \leq +\infty$, $\xi, \xi' \in L^p(\Omega, \mathcal{F}_T, P)$, and both g and g' satisfy (2A3) and (5A4). Let (y, z) and (y', z') be, respectively, the maximal (resp. minimal) L^p solution of $\text{BSDE}(\xi, g)$ and $\text{BSDE}(\xi', g')$ (recall [Corollary 5.1](#)). If $\xi \leq \xi'$ dP -a.s. and for each $(y, z) \in \mathbb{R} \times \mathbb{R}^d$,

$$g(t, y, z) \leq g'(t, y, z) \quad dP \times dt\text{-a.e.},$$

then for each $t \in [0, T]$, we have

$$y_t \leq y'_t \quad dP\text{-a.s.}$$

Finally, by virtue of [Theorems 5.1](#) and [2.1](#) we can obtain the following existence and uniqueness result for L^p solutions of BSDEs.

Theorem 5.5. Assume that $0 < T \leq +\infty$ and that g satisfies (2A1)–(2A3) and (3A2). Then for each $\xi \in L^p(\Omega, \mathcal{F}_T, P)$, $\text{BSDE}(\xi, g)$ has a unique L^p solution.

Proof. It follows from (2A1), (2A2) and (3A2) that $dP \times dt$ -a.e., for each $(y, z) \in \mathbb{R} \times \mathbb{R}^d$,

$$\begin{aligned} g(\omega, t, y, z) \operatorname{sgn}(y) &\leq u(t)\rho(|y|) + |g(t, 0, z) - g(t, 0, 0)| + |g(t, 0, 0)| \\ &\leq u(t)(k|y| + k) + v(t)(a|z| + b) + \bar{u}(t)\bar{\varphi}(0) \\ &= ku(t) + bv(t) + \bar{u}(t)\bar{\varphi}(0) + ku(t)|y| + av(t)|z|, \end{aligned} \quad (55)$$

where $k > 0$ is the constant of linear growth for the function ρ . Consequently, (5A1) with $f_t = ku(t) + bv(t) + \bar{u}(t)\bar{\varphi}(0) \in L^1([0, T]; \mathbb{R}_+)$ holds true for the generator g . Thus, the existence of an L^p solution of BSDE(ξ, g) follows from Theorem 5.1. Finally, the uniqueness follows directly from Theorem 2.1. The proof is then completed. $\square$

6. Existence and uniqueness of L^1 solutions

In this section, we will establish two existence results, two comparison theorems and an existence and uniqueness result for the (maximal and minimal) L^1 solutions of BSDEs. For convenience of expression, let us first fix a real number $\alpha \in (0, 1)$ and let $L^{\frac{2}{2-\alpha}}([0, T]; \mathbb{R}_+)$ denote the set of nonnegative functions $\lambda(\cdot)$ from $[0, T]$ to $\mathbb{R}_+$ such that

$$\int_0^T \lambda^{\frac{2}{2-\alpha}}(t) dt < +\infty.$$

In the sequel, let us introduce the following assumptions on the generator g , where $0 < T \leq +\infty$, and $a \wedge b$ represents the minimal number between a and b .

(6A1) There exist a nonnegative $(\mathcal{F}_t)$ -progressively measurable process $(f_t)_{t \in [0, T]} \in L^1([0, T] \times \Omega)$ and three functions $u(\cdot) \in L^1([0, T]; \mathbb{R}_+)$, $v(\cdot) \in L^2([0, T]; \mathbb{R}_+)$ and $\lambda(\cdot) \in L^{\frac{2}{2-\alpha}}([0, T]; \mathbb{R}_+)$ such that $dP \times dt$ -a.e.,

$$g(\omega, t, y, z) \operatorname{sgn}(y) \leq f_t(\omega) + u(t)|y| + (v(t)|z|) \wedge (\lambda(t)|z|^\alpha), \quad \forall y, z.$$

(6A2) There exist a nonnegative $(\mathcal{F}_t)$ -progressively measurable process $(\bar{f}_t)_{t \in [0, T]} \in L^1([0, T] \times \Omega)$ and three functions $\bar{u}(\cdot) \in L^1([0, T]; \mathbb{R}_+)$, $\bar{v}(\cdot) \in L^2([0, T]; \mathbb{R}_+)$ and $\bar{\lambda}(\cdot) \in L^{\frac{2}{2-\alpha}}([0, T]; \mathbb{R}_+)$ such that $dP \times dt$ -a.e.,

$$\forall y \leq 0, \forall z \in \mathbb{R}^d, \quad g(\omega, t, y, z) \leq \bar{f}_t(\omega) + \bar{u}(t)|y| + (\bar{v}(t)|z|) \wedge (\bar{\lambda}(t)|z|^\alpha).$$

(6A3) There exist a nonnegative $(\mathcal{F}_t)$ -progressively measurable process $(\bar{f}_t)_{t \in [0, T]} \in L^1([0, T] \times \Omega)$ and three functions $\bar{u}(\cdot) \in L^1([0, T]; \mathbb{R}_+)$, $\bar{v}(\cdot) \in L^2([0, T]; \mathbb{R}_+)$ and $\bar{\lambda}(\cdot) \in L^{\frac{2}{2-\alpha}}([0, T]; \mathbb{R}_+)$ such that $dP \times dt$ -a.e.,

$$\forall y \geq 0, \forall z \in \mathbb{R}^d, \quad -g(\omega, t, y, z) \leq \bar{f}_t(\omega) + \bar{u}(t)|y| + (\bar{v}(t)|z|) \wedge (\bar{\lambda}(t)|z|^\alpha).$$

(6A4) There exist a nonnegative $(\mathcal{F}_t)$ -progressively measurable process $(f_t)_{t \in [0, T]} \in L^1([0, T] \times \Omega)$ and three functions $u(\cdot) \in L^1([0, T]; \mathbb{R}_+)$, $v(\cdot) \in L^2([0, T]; \mathbb{R}_+)$ and $\lambda(\cdot) \in L^{\frac{2}{2-\alpha}}([0, T]; \mathbb{R}_+)$ such that $dP \times dt$ -a.e.,

$$|g(\omega, t, y, z)| \leq f_t(\omega) + u(t)|y| + (v(t)|z|) \wedge (\lambda(t)|z|^\alpha), \quad \forall y, z.$$

Remark 6.1. It is clear that $(6A4) \iff (6A1) + (6A2) + (6A3)$, and $(6A4) \implies (5A5)$.

The following lemma will be our basic tool in the treatment of L^1 -solutions.

Lemma 6.1. Assume that $0 < T \leq +\infty$, the generator g satisfies (44) and (2A5) (or (2A5')), and the process $g(t, 0, 0) \in L^1(\Omega \times [0, T])$. Then for each $\xi \in L^1(\Omega, \mathcal{F}_T, P)$, BSDE(ξ, g) has a unique L^1 solution.

Proof. The uniqueness follows from Theorem 2.4. Using a similar argument to Theorem 6.3 in Briand, Delyon, Hu, Pardoux and Stoica [5] and making use of Lemma 5.1 and (21), we can prove the existence. The proof is standard, we omit it here. $\square$

Similar to Lemma 5.2, we can prove the following Lemma 6.2.

Lemma 6.2. Assume that $0 < T \leq +\infty$, $\xi \in L^1(\Omega, \mathcal{F}_T, P)$, and the generator g satisfies (2A3) and (6A1). Let $(y_t, z_t)_{t \in [0, T]}$ be any L^1 solution of BSDE(ξ, g) and $(y'_t, z'_t)_{t \in [0, T]}$ the unique L^1 solution of BSDE($|\xi|, g'$) (recall Lemma 6.1), where

$$g'(\omega, t, y, z) := f_t(\omega) + u(t)|y| + (v(t)|z|) \wedge (\lambda(t)|z|^\alpha), \quad \forall \omega, t, y, z.$$

Then for each $t \in [0, T]$, we have

$$|y_t| \leq y'_t \quad dP\text{-a.s.}$$

Proof. We consider the function $\kappa(t, x) := (v(t)x) \wedge (\lambda(t)x^\alpha)$ for $(t, x) \in [0, T] \times \mathbb{R}_+$. It is not hard to verify that for each $t \in [0, T]$, $\kappa(t, \cdot)$ is nondecreasing and sub-additive on $\mathbb{R}_+$, i.e., $\kappa(t, x_1 + x_2) \leq \kappa(t, x_1) + \kappa(t, x_2)$ for each $x_1, x_2 \in \mathbb{R}_+$. Based on this fact, we can prove that for each $t \in [0, T]$ and $(z_1, z_2) \in \mathbb{R}^d \times \mathbb{R}^d$,

$$\begin{aligned} & |(v(t)|z_1|) \wedge (\lambda(t)|z_1|^\alpha) - (v(t)|z_2|) \wedge (\lambda(t)|z_2|^\alpha)| \\ & \leq (v(t)|z_1 - z_2|) \wedge (\lambda(t)|z_1 - z_2|^\alpha). \end{aligned} \quad (56)$$

Then we can see that g' satisfies (44) and (2A5'). Thus, note that $|\xi| \geq 0$ dP -a.s. and $g'(t, 0, 0) = f_t \geq 0$ $dP \times dt$ -a.e., Theorem 2.4 yields that for each $t \in [0, T]$,

$$y'_t \geq 0 \quad dP\text{-a.s.} \quad (57)$$

It follows from (6A1) that $dP \times dt$ -a.e.,

$$\forall y > 0, \forall z \in \mathbb{R}^d, \quad g(\omega, t, y, z) \leq g'(\omega, t, y, z). \quad (58)$$

Then, in view of (57), (58), Remark 2.2 and the fact that $\xi \leq |\xi|$ dP -a.s., by Theorem 2.4 we deduce that for each $t \in [0, T]$,

$$y_t \leq y'_t \quad dP\text{-a.s.}$$

Furthermore, by (6A1) we can also deduce that $dP \times dt$ -a.e.,

$$\forall y < 0, \forall z \in \mathbb{R}^d, \quad -g'(\omega, t, y, z) \leq g(\omega, t, y, z). \quad (59)$$

On the other hand, it is not difficult to verify that $(-y'_t, -z'_t)_{t \in [0, T]}$ is the unique L^1 solution of BSDE($-|\xi|, -g'$). Then, in view of (57), (59), Remark 2.2 and the fact that $-|\xi| \leq \xi$ dP -a.s., by Theorem 2.4 we can deduce that for each $t \in [0, T]$,

$$-y'_t \leq y_t \quad dP\text{-a.s.}$$

Thus, we have completed the proof of Lemma 6.2. $\square$

The following [Theorem 6.1](#) establishes an existence result on L^1 solutions of BSDEs, which is one of main results in this section. It improves considerably the corresponding existence results obtained in Briand and Hu [6] and Briand, Delyon, Hu, Pardoux and Stoica [5] for the one dimensional case, even for the case of finite time horizon.

Theorem 6.1. Assume that $0 < T \leq +\infty$ and g satisfies (2A3), (3A2) and (6A1) with $f_t \in L^1([0, T]; \mathbb{R}_+)$. Then for each $\xi \in L^1(\Omega, \mathcal{F}_T, P)$, BSDE(ξ, g) has an L^1 solution.

Proof. Let us first assume that ξ is nonnegative. Note that (6A1) $\implies$ (5A1). By [Theorem 3.1](#) and the proof of [Theorem 5.1](#) we know that BSDE(ξ_n, g) has a maximal bounded solution $(y_t^n, z_t^n)_{t \in [0, T]}$ for each $n \geq 1$, where $\xi_n := \xi \wedge n$. Furthermore, by [Corollary 3.1](#) we also know that for each $t \in [0, T]$, $(y_t^n)_{n=1}^{+\infty}$ is nondecreasing.

In the sequel, for each $(\omega, t, y, z) \in \Omega \times [0, T] \times \mathbb{R} \times \mathbb{R}^d$, define

$$g'(\omega, t, y, z) := f_t + u(t)|y| + (v(t)|z|) \wedge (\lambda(t)|z|^\alpha).$$

By (56) we know that g' satisfies (44) and (2A5'). It then follows from [Lemma 6.1](#) that for each $n \geq 1$, BSDE($|\xi_n|, g'$) has a unique L^1 solution $(^ny_t', ^nz_t')_{t \in [0, T]}$. Furthermore, in view of (6A1), by [Lemma 6.2](#) we can conclude that for each $t \in [0, T]$ and each $n \geq 1$, $|y_t^n| \leq ^ny_t'$ dP-a.s.

On the other hand, in view of $\xi \in L^1(\Omega, \mathcal{F}_T, P)$, it follows from [Lemma 6.1](#) that BSDE($|\xi|, g'$) has also a unique L^1 solution $(y_t', z_t')_{t \in [0, T]}$. Then, in view of $|\xi_n| \leq |\xi|$, by [Theorem 2.4](#) we know that for each $t \in [0, T]$ and each $n \geq 1$, $^ny_t' \leq y_t'$ dP-a.s.

As a result, we have proved that for each $t \in [0, T]$ and $n \geq 1$,

$$-y_t' \leq y_t^n \leq y_t^{n+1} \leq y_t' \quad \text{dP-a.s.} \quad (60)$$

In the sequel, arguing as in [Theorem 5.1](#), we define $y_\cdot = \lim_{n \rightarrow \infty} y_\cdot^n$ and use the localization procedure to find a process $z_\cdot$ such that $(y_\cdot, z_\cdot)$ is a solution of BSDE(ξ, g). Furthermore, by (60) we know that $y_\cdot$ belongs to the class (D) and the space $\mathcal{S}^\beta$ for each $\beta \in (0, 1)$, and then arguing as in Lemma 3.1 of Briand, Delyon, Hu, Pardoux and Stoica [5], in view of (6A1), we can deduce that $z_\cdot \in \mathcal{M}^\beta$ for each $\beta \in (0, 1)$. That is to say, $(y_\cdot, z_\cdot)$ is also an L^1 solution of BSDE(ξ, g).

In the general case, we can use a double approximation: $\xi_{n,p} := \xi^+ \wedge n - \xi^- \wedge p$ as in Briand and Hu [6] and Briand and Hu [7]. The proof is then complete. $\square$

The following [Theorem 6.2](#) gives a new existence result on the maximal and minimal L^1 solution of BSDEs.

Theorem 6.2. Assume that $0 < T \leq +\infty$ and g satisfies (2A3), (6A1) and (5A5). Assume further that g satisfies (6A2) (resp. (6A3)). Then for each $\xi \in L^1(\Omega, \mathcal{F}_T, P)$, BSDE(ξ, g) has a maximal (resp. minimal) L^1 solution.

Proof. We only prove the case of maximal solution, the other case is similar. Assume now that $\xi \in L^1(\Omega, \mathcal{F}_T, P)$ and that g satisfies (2A3), (6A1), (6A2) and (5A5). By (6A1) and (6A2) we know that dP $\times$ dt-a.e.,

$$g(\omega, t, y, z) \leq \tilde{f}_t(\omega) + \tilde{u}(t)|y| + (\tilde{v}(t)|z|) \wedge (\tilde{\lambda}(t)|z|^\alpha), \quad \forall y, z$$

where $\tilde{f}_t(\omega) := f_t(\omega) + \tilde{f}_t(\omega) \in L^1([0, T] \times \Omega)$, $\tilde{u}(t) := u(t) + \tilde{u}(t) \in L^1([0, T]; \mathbb{R}_+)$, $\tilde{v}(t) := v(t) + \tilde{v}(t) \in L^2([0, T]; \mathbb{R}_+)$ and $\tilde{\lambda}(t) := \lambda(t) + \tilde{\lambda}(t) \in L^{\frac{2}{2-\alpha}}([0, T]; \mathbb{R}_+)$. Then for

each $n \geq 1$ and $(\omega, t, y, z) \in \Omega \times [0, T] \times \mathbb{R} \times \mathbb{R}^d$, we can define

$$g_n(\omega, t, y, z) := \sup_{(u, v) \in \mathbb{R}^{1+d}} \{g(\omega, t, u, v) - n\tilde{u}(t)|y - u| - n[(\tilde{v}(t)|z - v|) \wedge (\tilde{\lambda}(t)|z - v|^\alpha)]\}. \quad (61)$$

In view of (56), arguing as in Fan, Jiang and Tian [19], we can also prove that the sequence of functions g_n is well defined for each $n \geq 1$, and it satisfies, $dP \times dt$ -a.e.,

- (i) $\forall y, z, g_n(\omega, t, y, z) \leq \tilde{f}_t(\omega) + \tilde{u}(t)|y| + (\tilde{v}(t)|z|) \wedge (\tilde{\lambda}(t)|z|^\alpha)$.
- (ii) $\forall y, z, g_n(\omega, t, y, z)$ non-increases in n .
- (iii) $\forall y_1, y_2, z_1, z_2$, we have

$$|g_n(\omega, t, y_1, z_1) - g_n(\omega, t, y_2, z_2)| \leq n\tilde{u}(t)|y_1 - y_2| + n[(\tilde{v}(t)|z_1 - z_2|) \wedge (\tilde{\lambda}(t)|z_1 - z_2|^\alpha)].$$

- (iv) If $(y_n, z_n) \rightarrow (y, z)$, then $g_n(\omega, t, y_n, z_n) \rightarrow g(\omega, t, y, z)$.

Furthermore, it follows from (61) and (6A1) that for each $n \geq 1$, $dP \times dt$ -a.e.,

$$\forall y < 0, \forall z \in \mathbb{R}^d, \quad g'(\omega, t, y, z) \leq g(\omega, t, y, z) \leq g_n(\omega, t, y, z), \quad (62)$$

where

$$g'(\omega, t, y, z) := -f_t(\omega) - u(t)|y| - (v(t)|z|) \wedge (\lambda(t)|z|^\alpha), \quad \forall \omega, t, y, z.$$

It follows from (56) that g' satisfies (44) and (2A5').

Note that for each $n \geq 1$, g_n satisfies (44) and (2A5') by (iii), and $|g_n(t, 0, 0)| \in L^1(\Omega \times [0, T])$ by (i) and (62). It follows from Lemma 6.1 that BSDE (ξ, g_n) has a unique L^1 solution $(y_t^n, z_t^n)_{t \in [0, T]}$ for each $n \geq 1$. In view of (ii) and (iii), by Theorem 2.4 we know that for each $t \in [0, T]$, the sequence $(y_t^n)_{n=1}^{+\infty}$ is non-increasing. On the other hand, by Lemma 6.1 we can let (y', z') be the unique L^1 solution of BSDE $(-|\xi|, g')$. Note that $y' \leq 0$ by Theorem 2.4. In view of (62), (iii), Remark 2.2 and the fact that $-|\xi| \leq \xi$ dP -a.s., by Theorem 2.4 we know that for each $t \in [0, T]$ and each $n \geq 1$,

$$y'_t \leq y_t^{n+1} \leq y_t^n \leq y_t^1 \quad dP\text{-a.s.} \quad (63)$$

In the sequel, arguing as in the proof of Theorems 5.2 and 6.1, in view of (63) and (iv), we can define $y = \lim_{n \rightarrow \infty} y_t^n$ and use the localization procedure to obtain a process z such that (y, z) is an L^1 solution of BSDE (ξ, g) .

Finally, it remains to show that (y, z) is the maximal one among all L^1 solutions of BSDE (ξ, g) . Indeed, let $(\hat{y}, \hat{z})$ be any L^1 solutions of BSDE (ξ, g) . In view of (iii) and the fact that $g_n \geq g$, by Theorem 2.4 we can obtain that for each $n \geq 1$ and each $t \in [0, T]$,

$$\hat{y}_t \leq y_t^n \quad dP\text{-a.s.}$$

Letting $n \rightarrow \infty$ yields the desired result. The proof is then completed. $\square$

The following corollary is an immediate consequence of Theorem 6.2 and Remark 6.1.

Corollary 6.1. Assume that $0 < T \leq +\infty$ and the generator g satisfies (2A3) and (6A4). Then for each $\xi \in L^1(\Omega, \mathcal{F}_T, P)$, BSDE (ξ, g) has both a maximal and a minimal L^1 solution.

Remark 6.2. A similar result to Corollary 6.1 has been obtained in the first version of Briand and Hu [6], where only the case of $0 < T < +\infty$ is considered, all of f_t , $u(t)$, $v(t)$ and $\lambda(t)$ are constants and we do not know whether the L^1 solution constructed by them is the maximal (or minimal) one or not.

Similar to Theorems 2.2 and 2.3, by virtue of Theorem 2.4 we can prove the following Theorems 6.3 and 6.4, which establish the comparison theorems on the maximal and minimal L^1 solutions of BSDEs under the assumptions of Theorem 6.2.

Theorem 6.3. Assume that $0 < T \leq +\infty$, $\xi, \xi' \in L^1(\Omega, \mathcal{F}_T, P)$, g and g' are two generators of BSDEs, and (y, z) is any L^1 solution of BSDE(ξ, g). Assume further that g' satisfies (2A3), (6A1), (6A2) and (5A5), and (y', z') is the maximal L^1 solution of BSDE(ξ', g') (recall Theorem 6.2). If $\xi \leq \xi'$ dP-a.s. and (9) holds true, then for each $t \in [0, T]$,

$$y_t \leq y'_t \quad \text{dP-a.s.}$$

Theorem 6.4. Assume that $0 < T \leq +\infty$, $\xi, \xi' \in L^1(\Omega, \mathcal{F}_T, P)$, g and g' are two generators of BSDEs, and (y', z') is any L^1 solution of BSDE(ξ', g'). Assume further that g satisfies (2A3), (6A1), (6A3) and (5A5), and (y, z) is the minimal L^1 solution of BSDE(ξ, g) (recall Theorem 6.2). If $\xi \leq \xi'$ dP-a.s. and (8) holds true, then for each $t \in [0, T]$,

$$y_t \leq y'_t \quad \text{dP-a.s.}$$

Corollary 6.2. Assume that $0 < T \leq +\infty$, $\xi, \xi' \in L^1(\Omega, \mathcal{F}_T, P)$, and both g and g' satisfy (2A3) and (6A4). Let (y, z) and (y', z') be, respectively, the maximal (resp. minimal) L^1 solution of BSDE(ξ, g) and BSDE(ξ', g') (recall Corollary 6.1). If $\xi \leq \xi'$ dP-a.s. and for each $(y, z) \in \mathbb{R} \times \mathbb{R}^d$,

$$g(t, y, z) \leq g'(t, y, z) \quad \text{dP} \times \text{dt-a.e.},$$

then for each $t \in [0, T]$, we have

$$y_t \leq y'_t \quad \text{dP-a.s.}$$

By virtue of Theorems 6.1 and 2.4 we can obtain the following new existence and uniqueness result for L^1 solutions.

Theorem 6.5. Assume that $0 < T \leq +\infty$ and that g satisfies (2A1)–(2A3) and (3A2). Assume further that g satisfies (2A5) with $f_t \in L^1([0, T]; \mathbb{R}_+)$ or (2A5'). Then for each $\xi \in L^1(\Omega, \mathcal{F}_T, P)$, BSDE(ξ, g) has a unique L^1 solution.

Proof. We only prove the case where g satisfies (2A1)–(2A3), (3A2) and (2A5) with $f_t \in L^1([0, T]; \mathbb{R}_+)$. The other case is similar. It follows from (2A1), (2A5) with $f_t \in L^1([0, T]; \mathbb{R}_+)$, and (3A2) that dP $\times$ dt-a.e., for each $(y, z) \in \mathbb{R} \times \mathbb{R}^d$,

$$\begin{aligned} g(\omega, t, y, z) \operatorname{sgn}(y) &\leq u(t)\rho(|y|) + |g(t, 0, z) - g(t, 0, 0)| + |g(t, 0, 0)| \\ &\leq u(t)(k|y| + k) + \lambda(t)(f_t + |z|)^\alpha + \bar{u}(t)\bar{\varphi}(0) \\ &\leq \tilde{f}_t + ku(t)|y| + \lambda(t)|z|^\alpha, \end{aligned} \quad (64)$$

where $\tilde{f}_t := ku(t) + \lambda(t)f_t^\alpha + \bar{u}(t)\bar{\varphi}(0) \in L^1([0, T]; \mathbb{R}_+)$ by Hölder inequality and the assumptions of $u(t)$, $\lambda(t)$, f_t and $\bar{u}(t)$, and $k > 0$ is the constant of linear growth for ρ . Thus,

combining (55) and (64) yields that (6A1) holds true for the generator g . Thus, the existence of an L^1 solution of BSDE(ξ, g) follows from Theorem 6.1. Finally, the uniqueness follows directly from Theorem 2.4. The proof is then completed. $\square$

At the end of this paper, let us give two examples to illustrate some applications of the previous theoretical results for two practice problems, where $0 < T < +\infty$.

Example 6.1. Recently, Shamarova, Ramos and Aguiar [35] shows an application of BSDE in gene expression. In case of one protein, the generator function $g(y) : \mathbb{R} \mapsto \mathbb{R}$ for the synthesis/degradation rate is

$$g(y) = R \frac{ay^2}{1 + ay^2} - \rho y,$$

where R, a, ρ are positive constants. It is clear that the generator g does not depend on z and is uniformly Lipschitz continuous in y . Hence, by Theorem 6.5 we know that for each $\xi \in L^1(\Omega, \mathcal{F}_T, P)$, BSDE(ξ, g) has a unique L^1 solution.

Example 6.2. In the well known Kreps–Porteus utility (see, for example, El Karoui, Peng and Quenez [14] for details), the generator $g(y) : \mathbb{R}_+ \mapsto \mathbb{R}$ is defined by

$$g(y) = \frac{\beta c^\rho - y^\rho}{\rho y^{\rho-1}} = \frac{\beta}{\rho} (c^\rho y^{1-\rho} - y),$$

where $0 \neq \rho \leq 1$, $0 \leq c$ and $0 \leq \beta$. Note that the generator g is only defined on $\mathbb{R}_+$. We have to consider the existence and uniqueness of nonnegative solutions for BSDE(ξ, g) when ξ is a nonnegative random variable. The following argument will be divided into three cases: $\rho = 1$, $0 < \rho < 1$ and $\rho < 0$.

- In case of $\rho = 1$, $g(y) = \beta(c - y)$, $y \geq 0$. We define a function $\bar{g}(y) : \mathbb{R} \mapsto \mathbb{R}$ such that $\bar{g}(y) = \beta c$ when $y < 0$ and $\bar{g}(y) = g(y)$ when $y \geq 0$. Then $\bar{g}(y)$ is uniformly Lipschitz continuous on $\mathbb{R}$, and by Theorem 6.5, BSDE($\xi, \bar{g}$) has a unique L^1 solution (y, z) for nonnegative random variable $\xi \in L^1(\Omega, \mathcal{F}_T, P)$. Furthermore, in view of $\bar{g}(0) = \beta c \geq 0$ and $\xi \geq 0$, it follows from Theorem 2.4 that $y \geq 0$. Thus, (y, z) is the nonnegative and unique L^1 solution of BSDE(ξ, g).
- In case of $0 < \rho < 1$, we define a function $\bar{g}(y) : \mathbb{R} \mapsto \mathbb{R}$ such that $\bar{g}(y) = 0$ when $y < 0$ and $\bar{g}(y) = g(y)$ when $y \geq 0$. Then $\bar{g}(y)$ is of linear growth on $\mathbb{R}$, and by Corollary 6.1, BSDE($\xi, \bar{g}$) has both a maximal L^1 solution (Y, Z) and a minimal L^1 solution (y, z) for nonnegative random variable $\xi \in L^1(\Omega, \mathcal{F}_T, P)$. Furthermore, we construct a function $\underline{g}(y) : \mathbb{R} \mapsto \mathbb{R}$ such that $\underline{g}(y) = \beta(c^\rho - 1)y/\rho$ when $0 \leq y \leq 1$ and $\underline{g}(y) = \bar{g}(y)$ when $y < 0$ and $y > 1$. Then $\underline{g}(y)$ is uniformly Lipschitz continuous on $\mathbb{R}$, and BSDE($0, \underline{g}$) has a unique L^1 solution $(0, 0)$. Now, in view of $\xi \geq 0$ and $\bar{g}(y) \geq \underline{g}(y)$ for each $y \in \mathbb{R}$, it follows from Corollary 6.2 that $y \geq 0$. Hence, (Y, Z) and (y, z) are, respectively, the maximal and minimal nonnegative L^1 solution of BSDE(ξ, g).
- In case of $\rho < 0$, we construct a function $\bar{g}(y) : \mathbb{R} \mapsto \mathbb{R}$ such that $\bar{g}(y) = 0$ when $y < 0$ and $\bar{g}(y) = g(y)$ when $y \geq 0$. It is not very hard to verify that $\bar{g}(y)$ satisfies (3A2) and (2A1) with $u(t) \equiv 0$ and $\rho(x) = -\beta x/\rho$. It then follows from Theorem 6.5 that BSDE($\xi, \bar{g}$) has a unique L^1 solution (y, z) for nonnegative random variable $\xi \in L^1(\Omega, \mathcal{F}_T, P)$. Furthermore, in view of $\bar{g}(0) = 0$ and $\xi \geq 0$, it follows from Theorem 2.4 that $y \geq 0$. Hence, we can conclude that (y, z) is the nonnegative and unique L^1 solution of BSDE(ξ, g).

Remark 6.3. By virtue of the results obtained in Sections 2–5 of this paper, the corresponding conclusions for nonnegative bounded solutions and L^p ($p > 1$) solutions of BSDE(ξ, g) in Examples 6.1 and 6.2 are also true.

Acknowledgments

The author would like to thank the two anonymous referees for their very careful reading and helpful comments.

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